

From critical crossings to quenched near-critical universality in Voronoi percolation

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Abstract

Taking Cardy’s crossing formula for critical Poisson–Voronoi percolation as an input, we prove a universal joint limit for the near-critical crossing thresholds of rational polygonal quadrilaterals under the monotone coupling. Conditional on the Poisson tessellation, the threshold law converges in probability over tessellations to the same law as on the triangular lattice. Each model is normalized by its own expected number of color-pivotal sites for a unit-square crossing.

1 Introduction

Critical crossing probabilities describe the large-scale geometry of planar percolation at one density. Near-critical percolation asks how that geometry changes when the density moves on the scale at which macroscopic crossings begin to respond. For a random tessellation, there is a further question: does a typical fixed geometry produce the same limiting process, after averaging only the colors? We prove such a transfer for Poisson–Voronoi percolation, taking its scalar critical Cardy formula as an explicit input.

Cardy’s conformal crossing formula [10], proved for critical site percolation on the triangular lattice by Smirnov [23], identified a central prediction of two-dimensional universality. Schramm’s introduction of SLE supplied a description of possible conformally invariant interface limits [20]. Camia and Newman established convergence of the triangular exploration path to chordal SLE_6 and of the full collection of critical interfaces [9, 8]; Smirnov and Werner determined the polychromatic arm exponents [24]. The quad-crossing framework of Schramm and Smirnov provides another way to retain joint macroscopic connectivity information [21]. These developments distinguish the information contained in one crossing probability from that in a coupled critical configuration.

The pivotal mechanism for changing density is classical. Kesten’s scaling relations [15] and Nolin’s systematic near-critical estimates [16] connect crossing derivatives, alternating four-arm events, and the width of the critical window. Garban, Pete, and Schramm constructed continuum pivotal measures and the near-critical and dynamical scaling limits on the triangular lattice [12, 13]. Their theory gives the principal continuum reference for this problem. Ahlberg and Steif, with an appendix by Pete, studied random thresholds under the canonical monotone coupling, including triangular crossing events [4]. Here the reference is specified directly through the triangular model and its own mean pivotal count; the proof below establishes the normalization in that convention.

For Voronoi percolation, Bollobás and Riordan proved that the critical coloring parameter is $1/2$ [7], and Tassion established box-crossing estimates at every scale [25]. Quenched estimates concern the color law after the Poisson geometry has been fixed. Benjamini, Kalai, and Schramm

posed the corresponding crossing-concentration question [6, Section 5.1]. Ahlberg, Griffiths, Morris, and Tassion proved crossing concentration and quenched noise sensitivity [3]; Ahlberg, de la Riva, and Griffiths obtained stronger concentration rates [2]. Ahlberg and Baldasso bounded the transition window polynomially [1]. Vanneuvillle's annealed scaling relations and quantitative quenched arm estimates [27, 29] supply the local control used here, particularly for independent color replicas sharing one tessellation.

Vanneuvillle's annealed spectral sample also identifies the crossing decorrelation scale for frozen-tessellation color dynamics and gives exceptional percolation times [28]. The present problem concerns a monotone coupling rather than color resampling and asks for its joint conditional threshold law. Our input is the scalar annealed Cardy conclusion of the companion manuscript [17, Theorem 1.1]. In particular, critical joint convergence, convergence of pivotal responses, and concentration of the near-critical conditional law are proved below rather than included in that input.

1.1 Models, crossings, and thresholds

A *quad* is a bounded Jordan domain $D \subset \mathbb{R}^2$ with four distinct boundary marks a, b, c, d in counterclockwise order. For a tiling by colored closed cells, let C_Q be the event that the union of the black cells, intersected with $\overline{D}$, contains a connected subset meeting both closed arcs ab and cd . Connections must remain in $\overline{D}$; no boundary wiring is used. Let Q_* be the unit square with left and right sides as its target arcs. Let $\mathcal{Q}_{\text{rat}} = \{Q_1, Q_2, \dots\}$ be one fixed enumeration of all quads with simple rational polygonal boundary and distinct rational boundary marks, including Q_* . The same enumeration and crossing convention will be used in both models.

For $\varepsilon > 0$, let η_ε be a full-plane homogeneous Poisson process of intensity ε^{-2} . Its closed Voronoi cells are

$$V_v = \{z \in \mathbb{R}^2 : |z - v| \leq |z - w| \text{ for every } w \in \eta_\varepsilon\}.$$

Give its sites independent uniform marks $U_v \in [0, 1]$, independently of the geometry. A site, and its cell, is black at density p exactly when $U_v \leq p$. The same geometry and marks are used at every density. The superscript V will denote this model.

For $\delta > 0$, the triangular sites are

$$T_\delta = \{\delta(k + le^{i\pi/3}) : k, l \in \mathbb{Z}\},$$

with their closed regular hexagonal Voronoi cells and independent uniform site marks. The superscript Δ denotes this model. Only the coloring is random here. A site is *color pivotal* for a crossing if flipping its color, while keeping every other color and the tiling fixed, changes the crossing indicator.

Let N_ε^V and N_δ^Δ count the color-pivotal sites for C_{Q_*} at density $1/2$. Define

$$r_\varepsilon^V = \frac{1}{\mathbb{E}N_\varepsilon^V}, \quad r_\delta^\Delta = \frac{1}{\mathbb{E}N_\delta^\Delta}. \quad (1)$$

The expectations include geometry in the Voronoi model and colors only in the triangular model. Their finiteness and positivity are proved in Lemma 2.12. Neither normalization depends on a sampled geometry. In either model, writing e for its mesh parameter and $M \in \{V, \Delta\}$ for its type, set

$$p_e^M(\lambda) = \max\{0, \min\{1, \frac{1}{2} + \lambda r_e^M\}\}, \quad \tau_e^M(Q) = \inf\{\lambda : C_Q \text{ occurs at } p_e^M(\lambda)\}. \quad (2)$$

The infimum of the empty set is $+\infty$, and an event occurring for every real parameter has threshold $-\infty$.

Put $\overline{\mathbb{R}} = [-\infty, +\infty]$ and $\theta(t) = \frac{2}{\pi} \arctan t$, with $\theta(\pm\infty) = \pm 1$. The threshold vector takes values in the compact metric space

$$\mathcal{K} = \overline{\mathbb{R}}^{\mathcal{Q}_{\text{rat}}}, \quad d_{\mathcal{K}}(s, t) = \sum_{j \geq 1} 2^{-j} |\theta(s(Q_j)) - \theta(t(Q_j))|. \quad (3)$$

For probability measures on $\mathcal{K}$ we use

$$d_{\text{BL}}(\mu, \nu) = \sup_{\substack{\|f\|_{\infty} \leq 1 \\ \text{Lip}_{d_{\mathcal{K}}}(f) \leq 1}} \left| \int f d\mu - \int f d\nu \right|.$$

This metric induces weak convergence. We write $\tau_e^M = (\tau_e^M(Q_j))_{j \geq 1}$.

1.2 The critical input and the main result

The following is the sole supplied Voronoi scaling-limit input.

Hypothesis 1.1 (Scalar critical Cardy formula). For every fixed bounded Jordan quad $Q = (D; a, b, c, d)$, the annealed probability of C_Q in fair-colored full-plane Poisson–Voronoi percolation converges, as $\varepsilon \downarrow 0$, to

$$F(x) = \frac{\int_0^x [u(1-u)]^{-2/3} du}{\int_0^1 [u(1-u)]^{-2/3} du}, \quad (4)$$

where a conformal map from D to the upper half-plane sends (a, b, c, d) to $(0, x, 1, \infty)$.

This is the conclusion of [17, Theorem 1.1] at precisely its scalar, geometry-averaged scope. We make the hypothesis explicit because the transfer proved here concerns much more information than that conclusion.

Theorem 1.2 (Quenched near-critical universality). *Assume Hypothesis 1.1. The expectations in (1) are finite and strictly positive for every mesh, and both scales tend to zero as their mesh tends to zero. There is a probability measure ν_{Δ} on $\mathcal{K}$ such that*

$$\text{Law}(\tau_{\delta}^{\Delta}) \implies \nu_{\Delta} \quad (\delta \downarrow 0). \quad (5)$$

For the independently normalized Voronoi model,

$$d_{\text{BL}}(\text{Law}(\tau_{\varepsilon}^V \mid \eta_{\varepsilon}), \nu_{\Delta}) \xrightarrow{\mathbb{P}}_{\varepsilon \downarrow 0} 0. \quad (6)$$

The conditional law averages all color marks and no geometry.

Equation (5) defines the reference entirely through the deterministic triangular model. Its construction and scale are independent of a Voronoi limiting object. We prove its existence in this normalization as part of the argument. The product topology records all finite collections of crossing thresholds, hence every finite collection of quads and near-critical parameters under the same monotone coupling. No coupling of different Poisson meshes is required for (6).

1.3 Proof strategy and transferable ingredients

There are two distinct transfers to make. First, one needs joint critical information from scalar crossings. We condition on a typical fixed geometry and explore only color bits. The critical arm estimates of Section 2 control contacts of the exploration and boundaries of unexplored domains. Cardy tests determine stopped hulls, whose iteration gives the usual SLE_6 exploration and loop limits in Section 3, extending the Camia–Newman exploration method to the geometries needed here. Section 4 then recovers crossings confined to the given polygons, including tests in which finitely many separated squares are assigned prescribed colors. Continuity under small changes of those regions and finite witnesses made of ordinary polygon crossings identify their common joint law.

Second, microscopic color changes must be compared with those fixed mesoscopic changes. For finitely many parameters, the law of one site’s entire bit history is its critical law plus a small centered signed measure. Multiplying these laws expands any finite crossing test over finite sets of sites. A forest of merging insertion locations selects disjoint critical arm probes. Its integrated bound decays faster than an exponential in the number of sites, so the expansion converges absolutely on every bounded parameter range (Section 5). Merging clusters and annulus structures have a methodological predecessor in the spectral-sample analysis of Garban, Pete, and Schramm [11, Section 4]; the coefficients here are signed insertion tests. No off-critical arm estimate is used in this expansion.

The comparison of its coefficients is local but must retain signs. We construct annular connections with a separating cycle consisting of four alternating monochromatic arcs. For exterior tests, the interior then supplies a single connection bit. Swapping the interiors of two independently sampled environments compares a single-site insertion with a square insertion, while preserving arbitrary signed history weights. Repeated trials in disjoint annuli make the error small relative to the required arm probabilities. An additional annulus protects a second color replica on the same geometry. Section 6 proves this comparison with its conditional probability and support requirements explicit.

The expansion is first normalized by the mean pivotal count for a fixed $2 : 1$ rectangle, for which the requisite pivotal estimate is available directly. Its first coefficient fixes the comparison constant without assuming convergence of a microscopic pivotal amplitude. Applying the same comparison to the unit square gives exactly (1). One- and two-replica convergence then yield concentration of the conditional law and complete the proof in Section 7. The forest bound and the signed local comparison isolate the two quantitative ingredients that may be useful in other transfers from critical to near-critical models.

2 Critical estimates and their geometric consequences

The proof uses critical estimates at two different levels. Uniform arm estimates, valid down to the mesh, control the expansion in sites. At fixed positive scales, the scalar Cardy hypothesis also gives quenched crossing limits, and these permit more precise planar constructions. We establish both levels here. In particular, the estimates for exploration near a boundary will be proved in conformal coordinates; a Euclidean half-plane estimate in the physical plane would not suffice for the domains produced by a stopped exploration.

2.1 Arm estimates, planar conventions, and localization

We suppress the model superscript when a statement applies to both models. The parameter e is the mesh, and the number of sites per unit area is

$$m_e = e^{-2} \quad \text{in the Poisson model,} \quad m_e = \frac{2}{\sqrt{3}e^2} \quad \text{on the triangular lattice.}$$

Fix a sufficiently large absolute constant B_0 and put $b_e = B_0 e$. Write $A_e(r, R)$ for the annealed probability of four alternating arms across a square annulus with inner and outer radii r, R . The center is a lattice site in the triangular model. Bounded changes of radii, and the replacement of square by circular annuli with margins, will be absorbed by quasi-multiplicativity. Arms of the same color are required to use disjoint sites, including the equal-colored neighboring arms in an odd arm sequence.

Lemma 2.1 (Uniform critical arm inputs). *For each model there are constants $C, c, \alpha, \beta > 0$ and $g \in (1, 2)$ such that, for $b_e \leq r \leq s \leq R$,*

$$c(r/R)^g \leq A_e(r, R) \leq C(r/R)^{1+\alpha}, \quad (7)$$

$$C^{-1}A_e(r, s)A_e(s, R) \leq A_e(r, R) \leq CA_e(r, s)A_e(s, R). \quad (8)$$

The one-arm probability is at most $C(r/R)^\beta$, and the polychromatic five-arm probability is at most $C(r/R)^2$. The alternating three-arm probability in a half-plane in a standard orientation is at most $C(r/R)^2$.

In the Poisson model, if $X_e(\eta; r, R)$ is the conditional probability of any of these specified arm events, then

$$(\mathbb{E}X_e(\eta; r, R)^2)^{1/2} \leq C \mathbb{P}(\text{the corresponding arm event}). \quad (9)$$

At fixed positive radii with $r \leq R/2$, the conditional arm probabilities minus their annealed probabilities converge to zero in probability over the geometry. Both assertions apply to the half-plane events just stated. For triangular four arms one may choose $g \in (5/4, 4/3)$.

These are critical inputs from the literature, with their precise scope as follows. For Poisson–Voronoi percolation, rescaling by e^{-1} gives unit intensity, independent fair colors, and inner radii at least one. The full-plane second-moment comparison and concentration are [29, Theorem 1.4]; the power estimates and quasi-multiplicativity are supplied by [29, Propositions 3.1–3.4 and 6.2]. The half-plane second-moment statement uses Remark 4.1 and equation (3.2) there. For its concentration assertion it is enough to use the weaker variance estimate proved in Section 4, equation (4.1). At unit intensity, this bounds the variance divided by the square of the annealed arm probability by Cr^{-h} , for some $h > 0$, $r \geq r_0$, and $r \leq R/2$. After rescaling, this relative variance is at most $C(e/r)^h$, with the same half-plane modification. No half-plane version of the sharper variance formula in Theorem 1.4 is needed.

The crossing and gluing method originates in the work of Russo and Seymour–Welsh [19, 22]. For the triangular lattice we use critical RSW, arm separation and quasi-multiplicativity as in [16, Sections 2.3 and 4], together with the arm exponents of [24]; see also [16, Theorems 21–22]. The two-scale inequalities with a strict exponent margin follow by iterating a sufficiently large fixed scale ratio and using quasi-multiplicativity. The finitely many remaining lattice scales change only the constant. The estimate for oblique straight boundaries needed below will be derived after Lemma 2.11; it is not being included implicitly in the standard-orientation input.

We use an embedded plane triangulation whose vertices are the sites. For a Voronoi tessellation, draw an edge between neighboring nuclei through the relative interior of their common cell side. Its faces may be curvilinear triangles. Almost surely the Poisson diagram is in general position. The same is true after finitely many fixed nuclei are inserted, outside a Lebesgue null set of insertion locations. All arguments take place in bounded regions and therefore involve locally finite diagrams. An interface, or *wall*, crosses each triangle through the edges whose endpoints have different colors. It can equally be drawn on cell boundaries without changing its order.

Site paths and paths in unions of cells can be interchanged with an error bounded by the sizes of the cells they use. In the interior of the plane, a same-colored turn at a triple vertex can be made through relative interiors of common sides. Thus a path with fixed positive clearance from the sides of a test can be drawn with clearance inside its color before being trimmed to that test. For alternating arms, the intervening opposite-colored paths keep the two paths of either color separate when this conversion is made in a buffered annulus.

We shall also approximate simple curves by graph paths. Here is the elementary approximation fact we use. In a thin open tube about a simple arc, follow the successive triangles met by the arc, take paths on their edges, and erase loops. Work first in a rectangular coordinate strip about the arc and let its width decrease more slowly than the mesh. The resulting paths converge in curve order: longitudinal backtracks tend to zero. For a closed curve, use an annular strip and retain the simple boundary facing the compact set to be enclosed. For crosscuts ending on a graph boundary, the same construction works up to their endpoints. A boundary cycle can first be made free of interior chords by retaining the part surrounding a fixed inner disk. Its inner row is connected along successive vertex stars, so crosscuts can reach the boundary without a bounded-degree assumption. Hex duality in these triangulated disks uses terminal sides, or uncolored fan terminals attached outside them.

Lemma 2.2 (Necessary arms for a local change). *Compare two colored tilings which agree outside a ball and throughout an intervening annulus. If their crossing indicators for a polygon quad differ, an annulus contained in the quad and surrounding the change has four alternating arms. Near one straight side, at scales larger than the distance to that side and smaller than the distance to all vertices and other sides, the corresponding half-annulus has three alternating arms. At a vertex or a marked point, a one-arm test from a neighborhood of that point to a fixed larger distance is always necessary. Each assertion allows fixed-factor margins in its inner and outer radii and uses only colors in the agreement region. Other prescribed local changes may be present, provided that they are held fixed in the comparison.*

Proof. Use an actual black closed-cell crossing in the configuration with a crossing. In the other configuration, closed-set planar duality gives an open white crossing in the complementary direction, relative to the quad. To see the applicable convention, map the quad to a rectangle and apply the separation theorem for a closed subset and its open complement. The white path can be chosen polygonal and with clearance from the black closed set. Both paths must visit the region where the configurations differ.

Trim each path at surrounding curves, using its first hit of the outer curve and its last preceding hit of the inner one. Its two parts lead to its two target sides. Outside the change, the two black parts cannot connect to each other: that would give the black crossing without the change. The same argument applies to the white parts, using their open clearance. Planar separation puts the four parts in alternating order. This proves the bulk implication.

Near one straight side, the parts leading to the other three sides give the three half-plane arms, with the middle one leading to the opposite side. The passage to disjoint site paths also respects

the literal closed-cell convention. A convex cell intersects a straight side in a connected set; a same-colored shared-cell shortcut either removes a repeated portion or would already connect the target sides without the change. For white arms use relative open paths and pass through cell interiors and common sides. Black paths of zero width on the straight boundary are represented by the cells containing them, with the allowed margins. At a corner, at least one target arc of the crossing is at a fixed positive distance; following the corresponding part of a crossing supplies the stated one-arm test. Nothing in this argument changes if other prescribed regions are held fixed: only the paths in the agreement annulus are used. $\square$

For Poisson geometry, agreement of the nuclei in a neighborhood alone does not immediately imply agreement of the cells there. The next lemma supplies the required screens and also explains how a failed screen is paid for in an arm estimate.

Lemma 2.3 (Buffered localization). *In a shell of radius $t \geq b_e$, one can choose a geometry event $\mathcal{L}_e(t)$, determined by the nuclei in a fixed enlargement of that shell, with*

$$\mathbb{P}(\mathcal{L}_e(t)^c) \leq C \exp[-c(t/e)^2]. \quad (10)$$

On this event the tiling and the vertex stars in a smaller shell are determined there, and every cell meeting that smaller shell has diameter and distance to its nucleus smaller than any prescribed fixed fraction of t . Fixed nuclei outside the enlarged shell do not affect this assertion. Screens with disjoint enlargements are independent; arm tests restricted by these screens and their color bits can likewise be used independently.

Suppose a comparison requires four arms wherever the screens pass between radii r, R . Its necessary probe, allowing screen failures, has probability at most $CA_e(r, R)$. The part in which a screen fails at a scale $t \geq Mb_e$ is at most

$$\delta(M)A_e(r, R), \quad \delta(M) \longrightarrow 0, \quad (11)$$

when the smaller starting scale is $r \asymp b_e$; the constants may depend on the bounded factors implicit here. For a fixed positive starting radius r , the relative failure contribution tends to zero as $e \rightarrow 0$. These conclusions also hold for probes consisting of four arms up to the distance from a straight side and three half-plane arms thereafter, once the half-plane estimate is bounded by CA_e .

Proof. Cover the enlarged shell by finitely many squares of side $c_1 t$, where c_1 is smaller than the required margins, and require a nucleus in each square. The Poisson empty-square formula and a union bound give (10). A point in the smaller shell then has a nucleus within $Cc_1 t$. A nucleus beyond the enlargement cannot be its closest nucleus. The same comparison along a segment in a convex cell bounds the entire part of any cell meeting the smaller shell and hence bounds its star. This is also why even a very long cell belonging to a remote prescribed insertion cannot cross a screened shell.

Use dyadic shells, with spare shells for the buffers. On complete success an ordinary arm event with slightly changed radii bounds the necessary probe. Otherwise group outcomes by the outermost failed screen, at scale t . Discard all demands up to scale Ct and keep the arm test beyond that scale, independently of the failed occupancy test. Quasi-multiplicativity and (7) give, with the convention that an inverted annulus costs one,

$$\exp[-c(t/e)^2]A_e(Ct, R) \leq CA_e(r, R)(t/r)^g \exp[-c(t/e)^2].$$

The dyadic sum is bounded uniformly. Its tail with $t \geq Mb_e$ goes to zero after division by $A_e(r, R)$ when $r \asymp b_e$. When r is fixed, the whole sum goes to zero. A failure at the outermost scale is

covered by the same inequality using the power lower bound for $A_e(r, R)$. This argument does not assert independence for events involving unrestricted Voronoi cells.

For a side probe, use bulk annuli up to the distance from the side and then half-annuli centered at the projection onto the side. Leave disjoint buffers where the two types meet. The assumed comparison of the half-plane cost with A_e , followed by quasi-multiplicativity, gives the same calculation. We shall verify the comparison in every triangular orientation below. $\square$

2.2 Quenched crossing limits and deterministic sequences

Write $\text{Cardy}(Q) = F(x_Q)$ for the scalar value in Hypothesis 1.1. The scalar Cardy hypothesis concerns an average over the geometry. We next remove that averaging for each fixed polygon test. The argument uses the squared arm estimates; ordinary annealed arm upper bounds alone would not yield the required variance decay.

Proposition 2.4 (Quenched scalar Cardy convergence). *For every fixed polygon quad Q ,*

$$\mathbb{P}^{\eta_e}(C_Q) \longrightarrow \text{Cardy}(Q) \quad \text{in probability over the Poisson geometry.}$$

The assertion holds simultaneously for every member of any fixed finite collection of polygon quads.

Proof. Let $Z_e = \mathbb{P}^{\eta_e}(C_Q)$. Restrict the relevant geometry to a large fixed box surrounding Q . The resulting error tends to zero exponentially as $e \rightarrow 0$ by localization. Partition this box into squares of side s , and resample the Poisson process in one square at a time. Denote the resulting conditional crossing probability by $Z_e^{(i)}$. Efron–Stein gives

$$\text{Var } Z_e \leq \frac{1}{2} \sum_i \mathbb{E}(Z_e - Z_e^{(i)})^2 + o_e(1). \quad (12)$$

Here and throughout this proof $e \rightarrow 0$ first, with $s > 0$ fixed. Couple the colors outside the resampled square. Screens make the two tilings agree outside a constant enlargement, except on an event tending to zero. A difference then requires the arm tests of Lemma 2.2 starting at radius Cs .

For squares a fixed distance from all vertices and marks, bulk and side tests, together with (9), contribute at most $Cs^{2+2\alpha}$ per square. If necessary decrease $\alpha > 0$ so that the side test has at least this exponent. There are $O(s^{-2})$ such squares.

We spell out the corner summation. Suppose the distance to the nearest vertex or mark, capped at a small fixed constant, is comparable to $u \gg s$. Probe from Cs to $c_Q u$, using four arms followed, when necessary, by half-plane arms. Choose c_Q small enough that the probe meets at most one side, also at narrow polygon angles. Independently, with a buffer, require one arm from $C_Q u$ to a fixed radius whenever the square is near a corner. The root-second-moment bound for the resulting necessary event is

$$C(s/u)^{1+\alpha} u^{\beta_0}$$

for some $\beta_0 > 0$. There are $O((u/s)^2)$ squares in this dyadic distance class. Their total contribution to (12) is at most $C(s/u)^{2\alpha} u^{2\beta_0}$. The bounded number of squares within distance Cs of a corner contribute $Cs^{2\beta_0}$ by the one-arm test alone. It follows that

$$\limsup_{e \rightarrow 0} \text{Var } Z_e \leq Cs^{2\alpha} + Cs^{2\beta_0} + C \sum_{\substack{u \text{ dyadic} \\ s \leq u \leq u_0}} (s/u)^{2\alpha} u^{2\beta_0} \longrightarrow 0$$

as $s \downarrow 0$. The dyadic sum tends to zero whether α is smaller than, equal to, or larger than β_0 ; in the equal case the extra factor is only $\log(1/s)$. The supplied scalar Cardy theorem identifies $\mathbb{E}Z_e$ and therefore proves the proposition. A finite union gives the last assertion. $\square$

Lemma 2.5 (Deterministic sequences and Jordan tests). *From every sequence $e \downarrow 0$ in the Poisson model one can extract a subsequence and realize its geometries on a common probability space such that, almost surely, the following hold. The mesh tends to zero on every compact set; crossing probabilities of all rational polygon quads tend to their Cardy values; and the fixed-positive-scale arm estimates in Lemma 2.1 hold for the conditional color laws, with a fixed constant slack, on any specified countable family of centers and radii. Conditional arm probabilities can also be compared to their mesh-dependent annealed values at these fixed scales.*

Any deterministic sequence satisfying these properties has the Cardy limit for Jordan quad approximations whose boundaries and marks converge in curve order to a Jordan quad. This includes graph-cycle approximations with fair interior vertices and terminal sides, and the literal closed-cell crossing convention. It also has uniformly positive crossing probabilities along any fixed finite collection of open tracks with compatible colors. The triangular sequence has all these properties deterministically.

Proof. Include the buffered enlargements used below in the countable annulus family before extracting. The first assertion is the subsequence principle for convergence in probability, followed by a diagonal extraction over compact sets, rational tests, and the specified annuli. Occupancy gives the compact mesh assertion; Proposition 2.4 and arm concentration give the others. At fixed radii the annealed probabilities are bounded away from zero by RSW and quasi-multiplicativity. For $r \leq R/2$, absolute concentration therefore gives the relative comparisons. For $r > R/2$, the arm event from r to $2R$ is contained in the event from r to R . Use this inclusion for the lower comparison and the bound 1 for the upper comparison; fixed-ratio RSW bounds the annealed probabilities above and below by positive constants. Nearby centers or radii are handled by inclusions with fixed-factor margins.

Here is the crossing-convention argument. A Schoenflies strip about each side of the limiting Jordan quad supplies inner and outer comparison quads. For a lower bound, trim in from the two non-target sides, also near their ends, and extend the target caps beyond the actual boundary. Keep positive clearance from the non-target sides. Approximate the comparison quad by a rational polygon. Any crossing of this polygon gives a crossing of the original test after trimming at its target sides. First convert the crossing to a path through cell interiors and common sides, using the clearance, and only then trim it. This order handles even a crossing initially supplied as a closed connected set.

The complementary color gives the opposite inequality by the same construction. As the strip widths decrease, both Cardy values tend to the desired value, by continuity of conformal maps under Jordan boundary convergence. For an approximating graph disk the comparison paths are trimmed on first meeting its boundary. Thus they use only fair interior values and reach neighbors of the terminal sides; prescribed exterior boundary colors are not used as parts of the crossing paths. Outer comparison tests similarly bound crossings originally expressed using closed cells. This proves the claimed sandwich for all the stated conventions.

For a fixed open track, use a finite chain of overlapping thin rectangles, together with transverse crossings in their overlaps. Their crossings concatenate, and such chains can also make circuits. Cardy or RSW gives a positive lower bound for each rectangle. Harris' inequality [14] gives the product lower bound for requirements of one color; opposite-colored requirements are placed in disjoint tracks and use disjoint sites once the mesh is small. For the triangular model start

with Smirnov's theorem [23] for the usual lattice approximations and apply exactly the same sandwich. Virtual terminal edges incur only a vanishing mesh error before the positive-clearance trimming. $\square$

In the rest of this section, a *deterministic sequence* means one satisfying Lemma 2.5. Probabilities then refer only to its independent fair colors. An assertion at fixed scales always means a limit superior along this sequence with those scales fixed first. This convention permits us to keep the Poisson geometry fixed throughout an exploration.

2.3 Color order and bulk estimates at fixed scales

The cited alternating arm estimates have to be strengthened before they can control arbitrary near contacts of paths. The required color-order argument is a finite-graph statement and therefore applies on every one of our deterministic tilings.

Lemma 2.6 (Color order in a rectangle). *Let a finite plane graph be drawn in a topological rectangle, with independent fair colors on its vertices. The probability of k vertex-disjoint crossings between two opposite sides, with a specified ordered list of colors, does not depend on that list. The assertion remains valid after arbitrary vertex deletions. Uncolored fan terminals on the two target sides may be shared; disjointness and order are then imposed on the paths and their incident edges off the fan terminals.*

Proof. Among the crossings of a prescribed first color, take the leftmost simple path. One way to make this choice precise is to add an external closing edge on the right. The union of all available crossing paths with that edge is a union of cycles sharing the edge and is two-connected. Its exterior envelope supplies the leftmost crossing. Equivalently, it has no bypass to its left between distinct path vertices, including the terminal incidences. Whether a proposed path is this leftmost path is determined by the colors on it and to its left; vertices strictly to its right remain independent fair bits.

If an ordered family exists, its first path may be replaced by this leftmost path while all remaining paths stay strictly to the right. Conditional on the leftmost path and the revealed left part, apply induction to the remaining graph. The conditional probability of its remaining $k - 1$ ordered crossings does not depend on their colors. Integrating proves independence of the last $k - 1$ colors. Global color reversal then removes dependence on the first one as well. Vertex deletions change neither the envelope construction nor the product law on the unrevealed vertices. $\square$

Lemma 2.7 (Bulk arms of arbitrary color order). *There are $c_0, C > 0$ such that, on a deterministic sequence and at fixed positive radii $r < R$, the probability of four disjoint bulk arms of any specified color order has limit superior at most $C(r/R)^{1+c_0}$. For six disjoint arms the corresponding bound is $C(r/R)^{2+c_0}$. The events may be taken between graph cycles in buffered circular or square annuli.*

Proof. We first compare a color sequence containing both colors with an alternating sequence. Choose two adjacent arms of opposite colors, trim them to proper crossings, and consider the gap between them. The disagreement interfaces in its triangulation contain a strand joining the two annular ends: the switch parity on the two ends, followed by pairing the interface ends, gives such a strand. Its visited triangles have no vertices on the other arms. Slitting along it and deleting the vertices of those triangles leaves a graph in the slit rectangle, possibly with further vertex deletions, containing the other arms. Each bank of the strand already supplies a monochromatic arm, of opposite colors on its two sides.

For precision, cut the actual disagreement curves at *every* contact with either annular boundary and sum over their maximal through components and their two color directions. Sum also over the at most k cyclic positions of the opposite-color gap; this fixes the residual ordered color list and costs only a factor depending on k . A candidate strand's occurrence is determined by the bits of its incident triangles. Conditional on those bits, the vertices left in the slit rectangle are still independent and fair. Lemma 2.6 replaces the ordered colors of the remaining arms by those completing an alternating family, without changing the conditional probability. The strand banks lie at the extremes relative to the remaining paths, so the replaced family and the banks really do give the alternating arm event.

The overcount in this sum has uniformly bounded moments. To see this, let N be the number of through strands after cutting at boundary contacts. In a smaller annulus, a bank of a wall gives a connected site walk: successive triangles connect their same-side vertices, with short spokes to the wall. The walk cannot cross another wall. After proper trimming it gives an arm in that gap. A gap is used by at most its two bordering strands, so N through strands imply at least a constant times N disjoint black arms in a buffered subannulus. If R/r is larger than a fixed constant, keep a fixed-ratio middle annulus. A white circuit there has probability at least $c > 0$ by Lemma 2.5; hence a black crossing has probability at most $1 - c$. BK on the fixed graph gives an exponential tail for the number of disjoint black crossings. Thus $\limsup \mathbb{E} N^p \leq C_p$ for every finite p , uniformly in R/r . Bounded ratios require only an enlarged constant.

Writing A_k^{alt} for the alternating event in the same annulus, with the fixed end margins just used absorbed in its radii, put $P_k^{\text{alt}} = \mathbb{P}(A_k^{\text{alt}})$. For each candidate strand, conditional color-order invariance replaces its residual-arm event by the completing alternating event with equal probability. Summing these replaced probabilities counts at most $C_k N$ strands on configurations with an alternating family. Therefore

$$\mathbb{P}(\text{specified mixed } k \text{ arms}) \leq C_k \mathbb{E}[N \mathbf{1}_{A_k^{\text{alt}}}]$$

The moment bound and Hölder's inequality give

$$\limsup \mathbb{P}(\text{specified mixed } k \text{ arms}) \leq C_p (\limsup P_k^{\text{alt}})^{1-1/p}.$$

For $k = 4$, choose p large in the upper bound $P_4^{\text{alt}} \leq C(r/R)^{1+\alpha}$. For $k = 6$, five of the alternating arms and the remaining arm have disjoint site witnesses. BKR for the fair product law and the five-arm and one-arm bounds give $P_6^{\text{alt}} \leq C(r/R)^{2+\beta}$. Another sufficiently large p preserves an exponent strictly larger than two.

Finally, an all-one-color k -arm event is a disjoint occurrence of the event of $k - 1$ such arms and the event of one such arm. Reimer's inequality at density $1/2$ bounds its probability by the intersection of the first event with the color-reversal of the second [18]. That intersection supplies a mixed k -arm family, since opposite colors are automatically site-disjoint. If color words are anchored, take the finite union over the cyclic position of the opposite-colored arm. The mixed-color estimate therefore proves the same bounds for monochromatic families. The bank and shadow argument used above is topological and also works after a homeomorphic change of coordinates with vanishing mesh. $\square$

2.4 Boundary coordinates, trimming, and extension

To control contacts with a boundary, we need an exponent greater than one for three disjoint arms using only fair interior vertices. The boundary of an unexplored domain need not be a simple

curve in the physical plane, so the estimate must be formulated in conformal coordinates. We first justify Cardy tests in those coordinates, then extend separated two-arm configurations to compare two nearby Cardy events.

Exploration produces graph disks whose physical boundaries may have several prime ends at the same point. We formulate precisely the coordinate setting in which Cardy and the boundary estimates will be used. Let D_n be graph disks, and let $f_n : \mathbb{D} \rightarrow \overline{D_n}$ be conformal in the open disk and continuous on its closure. Assume that f_n tends uniformly to f , that f is conformal in $\mathbb{D}$, and that its continuous boundary parametrization has no constant arc. The limiting boundary parametrization may identify different prime ends. We also allow a half-plane parameterization with its usual compactification, obtained by a Möbius change of disk coordinates.

Lemma 2.8 (Cardy tests in a boundary chart). *In the preceding setting the pulled-back triangulations have vanishing mesh. In a coordinate neighborhood of a compact open boundary interval on which f is one-to-one, Jordan tests with fair interior paths have their ordinary Cardy limits. Their paths may terminate at neighbors of boundary vertices, but no prescribed boundary colors are counted as parts of those paths.*

More generally, a test whose full boundary has a nonsimple physical projection has the same conclusion whenever each of its two complementary crossings admits cropped target arcs and a connecting subdisk with Jordan image, with the cropped Cardy values approaching the two complementary values of the original test. The cropping is performed before target arcs are extended for a comparison.

Proof. If a connected triangle piece had vanishing physical diameter but coordinate diameter bounded below, a Hausdorff subsequential limit of its preimage would be a nondegenerate continuum K on which f is constant. Uniform convergence of the maps gives this conclusion. Such a continuum cannot enter the open disk, where f is conformal and injective. A nondegenerate continuum in the boundary circle contains a nonconstant arc, contradicting the hypothesis. This proves the mesh assertion. Ordinary univalent distortion transfers the bulk estimates to balls in compact subsets of the coordinate disk.

In a patch where the indicated boundary interval projects injectively, crop the non-target sides of a coordinate test and trim its target arcs away from the ends. Its connecting subdisk has Jordan image. Approximate its interior sides by graph crosscuts and join them to the specified bank interval in order. The resulting cycles have Jordan boundary convergence in the physical plane. Apply Lemma 2.5, extending the target caps only after this cropping, and trim comparison paths at the graph boundary. They approach the specified bank from inside this subdisk, even if a different bank projects to the same physical neighborhood. Unused independent full-plane bits may be added for the comparison tests. Their trimmed witnesses use only the fair interior bits of the subdisk.

For the last assertion do this separately for the two complementary crossings. Their lower bounds approach the two complementary Cardy numbers; Hex duality supplies the squeeze. Thus the assertion invokes Cardy only for Jordan domains in the physical plane, not for a test on an unidentified covering surface or on the whole nonsimple projection. $\square$

We require an extension construction which does not condition on the colors exposed in finding arm endpoints. We therefore describe its retained event as an existence event with deterministic allowed regions. A *trim collar* is a fixed-ratio annulus, or half-annulus in a boundary chart, with a middle circle or semicircle and fixed-factor room on both sides. A proper arm traverses the collar from one boundary component to the other; after a trim it is stopped at its first hit from

the retained side. Small endpoint boxes will always be selected from a fixed finite family of box configurations on the middle curve.

Lemma 2.9 (Separated trims). *For every $z > 0$, one can choose a finite family of configurations of boxes on a trim curve, with fixed positive relative sizes and separations within each configuration, so that the following exception has limit superior probability at most z . The exception is that two distinct simple arms traversing a wider collar approach each other near the trim curve too closely for each retained endpoint to have its own box and a tenfold enlargement avoided by all the other arms. In a half-annulus also exclude visits too close to the two endpoints of the semicircle on the base. The conclusion applies to two arms in a half-annulus and to four arms, with fixed cyclic order, in a full annulus. It controls extra approaches of the arms to the endpoint boxes, as well as their first trim locations. The exception may be chosen as a union of local one-arm and four-arm tests in a fixed wider collar, whose width does not depend on z .*

Proof. We first record the boundary one-arm estimate, which does not use trimming. In a chart of Lemma 2.8, a half-annulus crossing has Cardy's limit. Its logarithmic image is a rectangle of aspect ratio $\pi^{-1} \log(R/r)$. The rectangle cross ratio in Cardy's formula gives a long-crossing bound $C \exp(-\pi t/3)$ at aspect ratio t , and hence

$$\limsup \mathbb{P}(\text{one free half-plane arm from } r \text{ to } R) \leq C(r/R)^{1/3}. \quad (13)$$

Margins give the same upper bound for graph cuts approximating the half-annulus. Only free interior vertices are used.

At either base endpoint of a trim semicircle, a visit within relative distance d by an arm continuing across the collar requires a one-arm event from scale d to a fixed scale. Equation (13) makes the combined probability arbitrarily small by first choosing d small. Away from these endpoint neighborhoods, cover the trim curve by $O(1/d')$ balls of relative radius $d' \ll d$. If two distinct arms approach in such a ball, their continuations in both directions give four disjoint site arms from scale Cd' to a fixed collar scale. Lemma 2.7 bounds the sum by $C_d(d')^{c_0}$, which tends to zero. Distortion is uniform on these balls because their radii are small compared with their distance from the coordinate boundary. In a full annulus there are no base endpoints, so only this second argument is needed.

Define the bad-collar event to be exactly the union of the endpoint one-arm tests and the four-arm tests on this grid. These tests are all contained in a fixed enlargement of the collar; making d, d' smaller does not enlarge that region. Choose a grid of boxes finer than the resulting separation tolerance, and take the finite family of all ordered selections with the required separations. Properly trim the simple arms between the chosen curves. Any approach of a different arm to an endpoint box would have triggered the same four-arm test: that arm and the arm ending there both continue out of the wider collar in both directions. Thus the tenfold enlargements may be reserved for their own arms alone. The retained portions use sites only on the retained side of the trim, up to a vanishing mesh error absorbed inside these boxes. Proper crossings preserve their linear or cyclic order. $\square$

Lemma 2.10 (Local guarded extension). *Fix the endpoint boxes and separations of Lemma 2.9, and prescribe separated destination intervals or patches on the other side of fixed-ratio extension regions, in the same order. In a full annulus a specified cyclic offset is allowed. Let H be the event that proper retained arms with specified colors exist between the trim curves, with endpoints in these boxes, and with every other arm avoiding each box's reserved enlargement. Require that each extension region overlap the retained region only in the corresponding reserved enlargement, where*

only the arm of its prescribed color is allowed. The arms can be extended to the destinations, with circuits or transverse crossings at their ends available for further connections, at a probability cost bounded below by a positive constant:

$$\mathbb{P}(H \text{ and the prescribed extensions}) \geq c\mathbb{P}(H). \quad (14)$$

The constant depends only on the normalized box and track layouts: it is uniform for similarity copies of a fixed finite family of layouts, including layouts in admissible coordinate charts. The bound holds for all sufficiently fine meshes of a deterministic sequence, with the mesh threshold allowed to depend on the fixed scales and chart. It remains valid with additional restrictions on bits outside the extension regions.

Proof. Around each endpoint box choose two slightly larger boxes. In the annular gap between them require a circuit of the endpoint's color, surrounding the endpoint box. The retained path reaches the small box from outside the larger one, so this circuit intersects it. Begin a corridor inside the circuit and require a crossing chain from there to the prescribed destination. This is the lasso construction: the circuit connects every possible retained witness to the corridor, without specifying where the witness first entered its endpoint box.

Route the corridor normally away from the trim until it is on the extension side with clearance, and then route it along a polygonal open track to its destination. The preserved order makes the tracks disjoint. Boxes can be made small compared with the track layout, or the tracks can taper near the boxes. Further lassos or transverse crossings may be placed at the destinations. Their rectangle chains have positive lower bounds by Lemma 2.5; boundary destinations use Lemma 2.8. Cardy's values depend only on the normalized coordinate shapes, so these lower bounds are uniform over the stipulated layouts and do not depend on the physical distortion of a chart. All tracks have positive widths. No path is required to follow a prescribed curve exactly.

Let U_B, U_W be the sets of sites used by the black and white extension requirements. These sets are disjoint for fine meshes. Where an extension meets a possible retained arm, it lies inside that arm's reserved enlargement; all arms of the other color avoid it. Consequently, after fixing the bits outside $U_B \cup U_W$, the existence event H is increasing in black bits on U_B and in white bits on U_W . The extension event has exactly the same coordinate monotonicities. Give the white coordinates the reversed order and apply Harris' inequality to this product space. The extension event itself depends only on these coordinates and has probability at least $c > 0$. Thus the conditional probability of its intersection with H is at least c times the conditional probability of H . Integration proves (14), also in the presence of restrictions outside these coordinates. A bounded finite number of choices of endpoint boxes is handled by summing, or by retaining a choice of maximal probability. At no point do we condition on an already selected arm or assert that its unexplored side has a fresh law. $\square$

Lemma 2.11 (Free boundary-arm bounds). *In the coordinate patches of Lemma 2.8, let $A_{j,n}^+(r, R)$ denote the probability of j disjoint arms across a half-annulus, of any specified color order, using only fair interior vertices. With fixed positive radii, constant-factor margins, and room within the patch,*

$$\begin{aligned} \limsup_n A_{1,n}^+(r, R) &\leq C(r/R)^{1/3}, & \limsup_n A_{2,n}^+(r, R) &\leq C(r/R), \\ \limsup_n A_{3,n}^+(r, R) &\leq C(r/R)^{4/3}. \end{aligned} \quad (15)$$

Base vertices of prescribed boundary color are excluded. The same convention applies to a crosscut at which an exploration is stopped before querying its vertices. In a full-plane tiling, arms allowed

to approach a straight or smooth test boundary are covered by slightly extending the coordinate half-plane, with the resulting errors absorbed at the inner radius.

Proof. The one-arm assertion is (13). By Lemma 2.6, applied to a graph half-annulus viewed as a rectangle, it suffices for two arms to consider opposite colors. This use is valid after deleting all base vertices; it requires no law for them.

First suppose the two arms have separated trims at scales comparable to r and R , with fixed-factor extension room. Lemma 2.10 extends their small ends to disjoint intervals on the base within distance Cr of each other, and their large ends to two prescribed separated arcs on a boundary at scale R . There are only finitely many endpoint-box and color-order choices. We claim that the extended event has probability at most Cr/R in the limit.

Choose a half-disk at scale R , with the two far arcs reserved for the two colors. Vary one marked point t on the base through an interval containing both near endpoints. Define E_t to be the black crossing from its far arc to the base target arc on the right of t . The other marks stay in fixed separated positions at scale R . Up to color reversal and reflection, the near endpoints are black then white from left to right, whereas the far black and white arcs are on the left and right respectively. When $t = t_-$ lies to the left of the short near interval, the black connection gives E_{t_-} . When $t = t_+$ lies to its right, the white connection is a complementary crossing and excludes E_{t_+} . These are nested events on the same fair bits, and $t_+ - t_- = O(r)$. Terminal-side Hex duality gives the same implication on the graph approximation. The ordinary Cardy limits of the two tests follow from Lemma 2.8. Their cross ratios stay in a compact subinterval of $(0, 1)$, and Cardy's formula is differentiable there, with derivative in the normalized moving mark bounded by a constant. Therefore

$$\limsup \mathbb{P}(\text{extended two-arm event}) \leq \lim_n [\mathbb{P}(E_{t_-}) - \mathbb{P}(E_{t_+})] \leq Cr/R.$$

Smooth boundary pieces flat on the relevant base interval avoid any corner issue in this comparison. The extension lower bound now proves the same estimate for the separated retained event.

It remains to bound the trimming exceptions without assuming the estimate being proved. Let $F(k)$ be the supremum of the limit superior of the two-arm probability over the admitted half-annuli of ratio 2^k , allowing graph cuts converging to their nominal boundaries. Fixed-factor changes of radii alter k by a bounded integer. Choose the two end collars with disjoint buffers. Each bad collar has probability at most z by Lemma 2.9. On that failure keep the necessary two-arm event in the remaining half-annulus, deleting at most k_0 end octaves, where k_0 is absolute. The bad-collar test and the remaining arms use disjoint sites. Thus, for sufficiently large k ,

$$F(k) \leq C_z 2^{-k} + 2zF(k - k_0). \quad (16)$$

Intermediate graph cuts converging to dyadic semicircles justify the same recurrence when the original ends are not exact circles. Choose z so that $2z2^{k_0} < 1/2$, and then choose D large enough for the finitely many initial ratios and for C_z . Induction in (16) gives $F(k) \leq D2^{-k}$. This proves the two-arm bound for all ratios, with the mesh limit always taken first.

Finally three disjoint paths contain disjoint witnesses for a two-arm event and a one-arm event. BKR and the first two estimates give the exponent $1 + 1/3 = 4/3$. Color order in the rectangle makes this valid for every prescribed three-color list. Small changes of the base or graph boundary are handled with margins. For an upper bound permitting paths right up to a discretized physical boundary, one can extend the fair tiling slightly beyond that boundary and use the same Jordan sandwiches; no prescribed boundary colors are thereby counted as free arms. $\square$

Two consequences complete the interface with the later sections. First, for an oblique straight side on the triangular lattice, Lemma 2.11 gives the three-arm estimate at fixed scales for every orientation. The estimate is uniform over rotations and translations: otherwise choose violating meshes, extract a convergent subsequence of angles and of the translation modulo a fundamental cell, and sandwich its tests between slightly enlarged tests of the limiting orientation. The fixed-scale estimate with margins contradicts the violation. Scale the lattice and iterate independent buffered blocks of one fixed, sufficiently large ratio. Above a sufficiently large lattice scale this yields any exponent strictly below $4/3$; omitted bounded lattice scales are absorbed in the constant. A bounded shift of the line absorbs the last lattice approximation error. Choose

$$5/4 < g_\Delta < \gamma_\Delta < 4/3.$$

The resulting half-plane three-arm bound $C(r/R)^{\gamma_\Delta}$ is at most $CA_e(r, R)$ by the triangular lower bound with exponent g_Δ . For Poisson geometry rotation invariance and the exponent two give the same comparison. This verifies, in every orientation, the side-probe comparison used in Lemma 2.3 and later in the site expansion. The common exponent $g < 2$ in that expansion may be enlarged to cover both models; it need not be the sharper g_Δ used in this comparison.

Second, at a fixed exceptional chart point, where the neighboring boundary arcs are only separately injective, a free arm coming from positive coordinate distance has probability tending to zero as its approach neighborhood shrinks. Indeed such a path has physical diameter bounded below: otherwise a sequence of its connected coordinate traces would give a nondegenerate continuum fiber of f , excluded in the proof of Lemma 2.8. Meanwhile the physical images of the shrinking inner neighborhoods shrink by continuity of f . The ordinary physical one-arm bound therefore proves the claim. This also treats endpoints of chart patches. These fixed-point estimates, the bulk bounds of Lemma 2.7, and the free boundary bounds of Lemma 2.11 are the estimates used to rule out degeneracies of the explorations in Section 3.

2.5 Pivotal normalization

We return to the original laws of the two models. In the Voronoi model, the expectations in this subsection average geometry as well as colors. The preceding estimates also give the deterministic normalization used when density is varied. We first work with a rectangle of aspect ratio two, for which the pivotal estimate is available directly; Section 7 will transfer the resulting normalization to the unit square in the theorem.

Lemma 2.12 (Finiteness and the auxiliary pivotal scale). *For every mesh, the expected number of color-pivotal sites for Q_* and for*

$$Q_0 = [-1, 1] \times [-1/2, 1/2]$$

with left-right target sides is finite and strictly positive. At criticality,

$$\mathbb{E}N_e(Q_0) \asymp m_e A_e(b_e, 1). \tag{17}$$

Consequently, with

$$a_e = A_e(b_e, 1), \quad q_e = \frac{1}{\mathbb{E}N_e(Q_0)}, \tag{18}$$

we have $q_e \asymp (m_e a_e)^{-1}$ and $q_e \rightarrow 0$.

Proof. Only finitely many cells meet a given compact rectangle almost surely. Their number has moments bounded by a polynomial in e^{-1} as $e \downarrow 0$: localize in successively larger boxes using the occupancy screens, and combine their exponential failure tails with the ordinary Poisson count moments. At any fixed mesh the same argument gives finite moments. Conditional on this finite set, the crossing indicator is a nonconstant monotone Boolean function: it is zero when all cells are white and one when all are black. Some edge of a Boolean path from the all-white state to the all-black state is therefore pivotal. Every coloring has positive fair-color probability, proving strict positivity of the conditional mean and hence of the mean. The triangular assertion is immediate from its finite number of relevant cells.

For Poisson–Voronoi percolation, [27, Lemma 6.1 and equation (6.1)] estimates the derivative of the crossing probability of a $2 : 1$ rectangle by $R^2 \alpha_{4,p}(R)$ at unit intensity. Its derivative is with respect to the color parameter, so Russo’s formula counts precisely the color pivotals used here. To include $p = 1/2$, which is an open endpoint in that statement, fix R and let $p \downarrow 1/2$. The annealed correlation length tends to infinity for the sufficiently small defining crossing constant used there. Localization and the integrable cell count justify continuity of the derivative and of the finite-scale arm probabilities. Rescaling now gives (17). The usual rectangle event agrees with our intersection-with-the-rectangle convention, apart from null general-position exceptions.

On the triangular lattice, Kesten’s pivotal/four-arm method [15, Section 4], together with arm separation and extension as in [16], yields the same estimate. We verify the exact boundary convention directly. For the lower bound, every site in a small fixed central patch can send four separated alternating arms from distinct neighbors to macroscopic landing intervals directed toward the four sides. Inner arm separation, followed by a finite prescription at a bounded lattice radius, connects to the neighbors without using the site itself. Outer landing rectangles and disjoint RSW tracks continue the arms across the four side arcs away from corners. The probability is at least ca_e . A black setting makes the black crossing; a white setting makes an open white crossing with clearance, so this is pivotal for the stated closed-cell event. For the upper bound, use the straight-side three-arm estimate in every orientation established above. Write its exponent as $\gamma > g$, choosing the triangular $g < 4/3$. For a relevant lattice site let u be its distance to the nearest rectangle corner, capped at a small constant and bounded below by b_e , and let $d \leq u$ be its distance to the nearest side, also bounded below by b_e . Lemma 2.2 and independent buffered probes give

$$\mathbb{P}(\text{the site is pivotal}) \leq CA_e(b_e, d)(d/u)^\gamma \leq Ca_e d^{-g}(d/u)^\gamma.$$

When $d \asymp u$ the half-annulus is omitted. The number of sites in a dyadic (u, d) class is at most $Cm_e u d$, including the bounded-width row just outside the rectangle whose hexagons meet it. Summing first over dyadic $d \leq u$ gives at most $Cm_e a_e u^{2-g}$; summing over u is bounded since $g < 2$. The bounded number of corner-scale sites is covered by the same estimate and the lower bound $a_e \geq cb_e^g$. This proves the upper estimate with the exact closed-cell convention. Finally (7) gives $m_e a_e \geq ce^{g-2} \rightarrow \infty$, proving the last assertion. $\square$

3 Critical explorations and oriented loops

The scalar limit established in Proposition 2.4 does not determine the joint law of crossings. Our first step towards that law is to identify the critical interfaces, while keeping the tessellation fixed. The argument has three stages. First we identify the law of an exploration stopped on a crosscut, including its unexplored domain. We then recover the ordered path from these stopped domains. Finally we explore inside the remaining faces to obtain the oriented interface loops. Section 4 will

read the crossing tests needed for the near-critical argument from this common loop law.

Throughout this section we work along a deterministic sequence of tessellations supplied by Lemma 2.5. Thus mesh sizes on compact sets tend to zero, fixed polygon crossing probabilities converge to Cardy's formula, and the fixed-scale conclusions of Lemmas 2.7 and 2.11 hold. All probabilities in the section concern independent fair colors on these fixed tessellations. When boundary values are prescribed, only the interior colors are random. Every arm estimate is used with the mesh limit first and the radii fixed; subsequently the inner radius may decrease. In particular, no estimate conditioned on an arbitrary already explored color configuration is being asserted.

3.1 Exploration on a fixed triangulation

We use the embedded triangulation of sites described in Section 2. A graph disk is the closed region inside a simple site cycle, together with the triangles it contains. We choose the cycle without chords in its interior. Two specified boundary edges are switches: the boundary sites between them on one arc are black, and those on the other arc are white. The interface is drawn through each triangle between its black and white corners. It starts at one switch and ends at the other. At each step it queries an as yet unknown site only if that site's color is needed to choose the next edge. Equivalent drawings on Voronoi edges differ by a quantity tending to zero with the mesh.

Curves are considered modulo increasing reparametrization, with the uniform distance between representatives; constant pauses are irrelevant. Boundary convergence in order means convergence in this curve metric with the cyclic orientation retained. A sequence of graph disks approximates a Jordan domain if its boundary cycles converge in order to the Jordan boundary and its marked switch edges converge to the specified distinct marks.

Proposition 3.1 (Chordal exploration convergence). *Let graph disks on a deterministic sequence as above approximate a bounded Jordan domain D , with switches converging to distinct boundary points a, b . Give their interiors independent fair colors and their two boundary arcs the prescribed opposite colors. The exploration from a to b converges in law, in the curve metric, to chordal SLE_6 in D from a to b . This conclusion holds for every such converging sequence of disks and marks.*

We prove the proposition in the present and next two subsections. The hull identification follows the method of Camia and Newman [9]; the graph, boundary, and conditional-law arguments below establish its applicability here. The convergence theorem in [9] is not used as a theorem about the present triangulations.

The first observation explains why exploration can be continued without resampling the geometry or changing the conditional color law. Include the original boundary cycle, every vertex in the band of triangles visited by the exploration, and every edge between included vertices in the revealed graph. Unqueried sites outside this graph retain their independent fair colors, conditional on the exploration. Indeed an exploration history specifies only the queried colors; the decision to query a site is a function of the preceding answers and the fixed triangulation. Summing over histories with the same revealed data does not impose any additional constraint on unqueried coordinates.

Lemma 3.2 (Faces of the revealed band). *The revealed graph is two-connected. Each of its faces that contains unqueried sites has a simple, internally chord-free site cycle as boundary. For a nonempty explored band, such a cycle shares at most one interval with the original boundary cycle; the interval may be empty or a single vertex. After a completed exploration every such face has a*

monochromatic boundary. If an exploration is stopped with its outgoing edge facing the unqueried face containing the target switch, that face has exactly two switches, namely the target and the outgoing edge.

Proof. Begin with the boundary cycle. Adding the successive triangles of the exploration band preserves two-connectivity: a triangle is attached along the edge through which it is entered, and any new corner has two neighbors in the graph already present. Adding the remaining edges between revealed vertices preserves this property. Boundaries of bounded faces of a two-connected plane graph are simple cycles. An interior chord would be an edge between revealed vertices and would already have split the face, proving the chord assertion.

The part of the visited band off the original boundary is connected. Two successive steps have an interior corner in common whenever they meet the boundary: otherwise the common edge would be an interior chord of the initial cycle. The initial and final switch triangles cause no exception to this connectivity. If a face had two separate intervals on the original boundary, a crosscut through that face between two of its boundary contacts would leave parts of this connected interior band on both sides. They could not be joined without crossing the crosscut or the face. This proves the interval assertion, including an isolated contact vertex by taking a small crosscut around that vertex.

Draw the exploration as a simple dividing arc in the triangulated disk. Its black neighbors are on one bank and its white neighbors on the other; an edge joining the two banks belongs to the visited band. Consequently a complementary face sees only one color on its boundary. For the target face at a stop, continue the dividing arc through that face to the target. The same separation gives one black and one white boundary arc, meeting at the stated edges. $\square$

We next establish the compactness and contact properties needed both for stopped explorations and for recursion in the faces. Work first in round disk coordinates. For initial Jordan approximations their conformal maps converge uniformly on the closed disk. The coordinate mesh tends to zero by Lemma 2.8: a sequence of connected triangles with physical diameter tending to zero and coordinate diameter bounded below would yield a nondegenerate continuum on which the limiting map is constant. The same observation will apply to the maps of residual domains once their uniform convergence has been proved.

Lemma 3.3 (Tightness of paths and face boundaries). *Exploration paths are tight in the curve metric. The same holds jointly for boundaries of faces produced by a completed exploration and for the two colored arcs of a stopped target face. At any fixed positive scale their numbers of annular traversals are tight. These statements hold in the disk coordinates just described and, by uniform continuity of the conformal maps, in physical coordinates.*

Proof. We use the multiple-crossing compactness argument of Aizenman and Burchard [5], spelling out its fixed-scale application to these paths and face boundaries. Cut a wall traversal between two graph cycles in a buffered annulus. Following the corners of the successive triangles on either bank gives a connected site path, its *shadow*. Crop the shadow farther from the two ends so that it cannot run around an end of the wall traversal. Between disjoint proper wall traversals the corresponding shadows lie in the intervening gaps. Thus many traversals supply many site-disjoint arms, with at most a fixed loss at the ends. A positive fraction of these arms have one color. The BK inequality [26, Theorem 3.3], together with the positive probability of an opposite-color circuit in a fixed-ratio subannulus, bounds the probability of arbitrarily many such arms by a geometric tail. Increasing the annulus ratio if necessary makes the one-crossing bound strictly less than one.

In a boundary half-annulus, k disjoint proper wall traversals have at least $k - 1$ free shadows, in their internal gaps. Here and below *free* means that the arm uses only random interior sites. The original boundary sites can occur only in the two outside gaps. If neither bank of any traversal uses those sites, the two outside shadows are free as well, giving $k + 1$ free arms. These assertions can be read between intermediate graph crosscuts and then cropped, so that a large site star cannot join around a cut. They depend only on the plane triangulation. Lemma 2.11 and the same BK argument give the boundary estimates. For a face boundary the random pieces are monochromatic simple site paths. Its original-boundary pieces and switches are bounded in number and cause no additional oscillations. The free pieces obey the preceding arm bounds. Near each fixed switch use the fixed-point one-arm bound instead.

For clarity, fixed-annulus estimates suffice for tightness in the curve metric. Fix an excursion size $r > 0$. Cover first the switches, then the boundary, and then the remaining compact interior by finitely many smaller neighborhoods whose surrounding annuli have outer diameter less than $r/10$. Every successive displacement of size r requires a traversal of one of these annuli. Their crossing counts therefore bound the number of such displacements. Do this for $r = 2^{-j}$. Choose bounds with exceptional probabilities summable in j , and parametrize curves by assigning summably small time budgets to their successive steps at each scale. This gives a common modulus of continuity outside an arbitrarily small exceptional event. The same bounds apply simultaneously to face boundaries, since their crossings are among the site arms just counted. Finite initial mesh indices can be accommodated separately. The nondegeneration argument in Lemma 3.5 below supplies the finite numbers of macroscopic faces needed to list those boundaries jointly. $\square$

Lemma 3.4 (Contact rules). *In a subsequential curve limit, the exploration has no interior triple point, no double point on the original boundary, and no second visit to either switch. Its interior double points avoid every fixed line segment. It has no nonconstant interval lying on the boundary. Moreover a discrete excursion whose bank shadows avoid a specified boundary bank cannot acquire an intermediate contact with that bank in the limit, provided the positions immediately before and after that contact make positive excursions from it.*

For a random monochromatic arc of a face boundary, distinct interior visits to the same point are excluded. On an original boundary interval of its own color it cannot have an intermediate contact between positive excursions. A face of positive limiting diameter cannot attach to a boundary interval of its own color through an interval collapsing to a point, including a single-vertex attachment. These statements also hold with the colors reversed.

Proof. We first identify the arms forced by a contact. All traversals below are properly trimmed in a buffered annulus, so that their shadows cannot join around an annular end.

Exploration contacts. A triple visit in the bulk, with positive excursions between visits, gives six wall traversals and hence six free arms. A double visit on the original boundary gives four traversals and three free arms. A double visit on a fixed interior segment gives four free arms. If a wall comes to a boundary bank and leaves it without either shadow using that bank, the two traversals have three free shadows. A shadow uses a bank site only when a crossed edge is incident with that site, up to one triangle, so this count is independent of the precise wall drawing. A second visit to a specified switch requires a free arm from a vanishing neighborhood of that point. A nonconstant interval of the wall on the boundary likewise gives such an arm at some point of a fixed countable dense set of boundary points.

The exterior of a face. A monochromatic face arc needs an additional planar observation. At a finite mesh let C be the simple site cycle bounding the face. Every site on its random part has an opposite-colored neighbor across an edge of the explored band. These neighbors belong to one

connected opposite-colored bank, denoted by K , which includes its original boundary arc and lies in the closed exterior of C . Suppose four properly trimmed traversals of the random part of C cross a bulk annulus. Each traversal has an exterior-side incidence with a neighboring gap, and one gap accounts for at most two of these incidences. Thus at least two distinct gaps about a traversal on its exterior side. Other pieces of C may still enter such a gap. We show that each selected gap contains an opposite-colored radial crossing.

Choose one of these gaps between two proper traversals P, Q of a wider annulus, with the gap on the exterior side of P . Cut across this gap at two intermediate graph curves, leaving the ends of P, Q beyond the cuts. The resulting working quadrilateral has subarcs of P, Q as its lateral sides, and its interior avoids the whole of P and Q ; this construction allows radial backtracks. Write p_-, p_+ for the ends of the contiguous face-cycle subarc P , and join their opposite neighbors to them by short spokes beyond the working cuts. If K has no longitudinal crossing of the quadrilateral, rectangle separation supplies a transverse crosscut avoiding K , from an interior point x of P to Q , with interior disjoint from all of P . Stop it at its first subsequent meeting y with the entire cycle C . It starts on the exterior side of P , so this prefix lies in the exterior. Other pieces of C may cause an earlier stop, but y is outside P because the original crosscut avoids P .

The relevant ambient region is the whole exterior of C , compactified by the point at infinity: it is a topological disk. The boundary points occur in the cyclic order p_-, x, p_+, y . Our crosscut therefore separates the two endpoint spokes in this disk. The spokes avoid its working annulus and put their opposite-colored endpoints in different components. Those endpoints cannot be joined by K without crossing the crosscut, a contradiction. In particular a connection going around an end of the annulus does not escape this separation. Crop once more to remove the short spokes. The resulting opposite-colored crossings in the two selected gaps are disjoint; with the four monochromatic traversals they supply six free arms.

A self-return of the random face arc in the bulk forces exactly this configuration. Near the original boundary, four free traversals already exceed the three-arm requirement. If the face arc has the same color as the nearby original bank, an intermediate contact gives two arms along the arc and one opposite arm from K . The opposite arm cannot use that original bank, so all three arms are free. The same count applies when the two ends of a macroscopic face arc approach each other as its shared boundary interval collapses; a singleton attachment is included. At a fixed switch or fixed patch endpoint, one free arm suffices. Contacts with an original bank of the other color need not be excluded: they will be retained as distinct boundary accesses.

Applying the arm estimates. For a fixed positive outer excursion scale, cover the compact interior by $O(r^{-2})$ balls of radius r . The six-arm bound in Lemma 2.7, whose exponent is greater than two, excludes both interior triple visits of the wall and self-returns of a face arc. Along a boundary interval there are only $O(r^{-1})$ balls, and the free three-arm exponent $4/3 > 1$ in Lemma 2.11 excludes the boundary configurations just listed. The four-arm exponent greater than one similarly excludes double visits on a fixed interior segment. The fixed-point one-arm estimate handles switches, patch endpoints, and the dense boundary tests.

All centers and outer radii are fixed before passing to the mesh limit. A contact at a random point with positive excursions is covered by an enlarged ball of a rational grid. Exhaust excursion sizes by a countable list and then decrease the grid scale. Thus none of these exclusions requires an arm estimate conditioned at a random limiting contact point. $\square$

Lemma 3.5 (Convergence of the remaining faces). *After a completed exploration, every non-constant subsequential limit of a remaining face boundary is a Jordan curve. Its interior is a component of the complement of the limiting exploration in the initial domain. Faces of diameter*

at least any fixed $r > 0$ have tight numbers, converge without surplus faces, and retain their colors and their intervals of attachment to the original boundary. Thresholds on diameters may be chosen outside the atoms of the limiting sizes.

Proof. A face boundary consists of its random part and at most one interval of the initial boundary, by Lemma 3.2. The random part has no self-return, by Lemma 3.4. If there is an original boundary interval, it has the face's color. The same lemma excludes an intermediate return to that interval or an additional visit to its endpoints. A nontrivial interval cannot have the entire free arc collapse, because its two endpoints are separated in the initial Jordan boundary parametrization; intervals under consideration contain no switch in their interior. Nor can a macroscopic free part have its attachment interval collapse. If no original interval is present, a nonconstant limiting cycle cannot fold into a doubly traced arc: that would be a distinct visit of its random part. After deleting pauses the limit is therefore Jordan.

These facts also give the claimed finiteness. If arbitrarily many disjoint faces had diameter at least r , their inscribed radii would have a subsequence tending to zero. Curve tightness would give a nonconstant limiting boundary of such a subsequence. Its Jordan interior contains a disk; convergence in order forces the approximating face to contain a slightly smaller disk, a contradiction. The argument proves tightness of the number and excludes surplus macroscopic faces in any coupled subsequential limit.

The face boundaries lie within a vanishing distance of the band or the initial boundary. Jordan convergence prevents the band from entering a limiting face interior. Conversely a compact ball in an open component off the limiting trace, together with finitely many compact joining paths in that component, eventually avoids the discrete band. It is thus recovered by one of the discrete faces. This identifies the components. The side of a fixed point off the trace is preserved by the winding number of the wall completed along the initial boundary, so the color labels pass to the limit. Finally a nontrivial original boundary interval cannot be supplied entirely by the limiting free arc: this would contradict the fixed-point one-arm test on that interval. Hence true intervals of attachment, as well as their order, are recovered. $\square$

3.2 Stopping on a crosscut and identifying the residual domain

Fix a smooth crosscut L in the current disk coordinates, separating the start from the target. We call its two complementary portions the front and the back, with the start in the front. The cuts used for iteration will be images of semicircles centered at 0 in the upper half-plane with marks $0, \infty$. Approximate L by a simple graph crosscut whose intermediate vertices are interior. Stop the exploration *before* it queries a vertex of that crosscut. If it passes the crosscut at an endpoint along boundary triangles, stop at the first triangle in that endpoint star instead. Such endpoint stops will have vanishing probability when the endpoint neighborhoods are shrunk after the mesh limit.

Lemma 3.6 (The stopped target face). *At a stop separated from the two endpoints of L , the outgoing edge faces the unqueried component containing the target. In a coupled subsequential limit its boundary consists of two individually simple colored banks, joining the new tip to the target, in prime-end order. In the current coordinates the target domain is the union of the back, two Jordan side domains in the front, and their two open attaching intervals on L . The two colored side arcs meet L only at their prescribed endpoints, one of which is the hit point. Normalized conformal maps onto the new target domains converge uniformly on their closed parameter disks. The same is true of the composed maps into the original physical domain.*

Proof. Until the stopping query, every interior vertex of the separating path is unknown. Its bulk portion joins, through unknown vertices on the back side, the target edge. The chord-free initial boundary ensures a connected inner row at that edge: the triangles adjacent to consecutive boundary edges meet through their interior corner stars. Consequently the outgoing edge is incident to the target face. This argument is applied outside fixed endpoint neighborhoods; chords from the crosscut to the rim near its ends cannot affect it. Lemma 3.2 gives the two colored boundary arcs of that face.

With probability tending to one as endpoint neighborhoods decrease, the portions of the original banks from the target up to, and a little beyond, the endpoints of L remain untouched. To reach one of these fixed neighborhoods, the band's bank of the opposite color must supply a free arm out of it. The fixed-point one-arm bound therefore applies. Each target-face arc consists of an original bank followed by random sites of the same color. Away from the termination, its simplicity follows from Lemma 3.4. There is one additional check at the separator: an extra approach of a random arc to L , with positive excursions of that arc before and after it, supplies two arms along the arc and an opposite arm through its neighbors in the band. All three arms lie on the front side of L , up to the vanishing mesh error. Lemma 2.11, applied in this smooth crosscut patch, excludes such an approach. The same count excludes an additional visit to the termination. The fixed endpoint tests handle the two ends of L .

The residual face is now described without identifying distinct boundary accesses. In the discrete target face the separator goes from one old bank to the other, and the queried vertex would be next to the endpoints of the outgoing edge. Cutting there, and removing that one triangle if necessary, separates two side polygons from the back. In the limit each side polygon has a boundary formed by a subinterval of its own old bank, its simple random arc, and its subarc of L . This is a Jordan curve by the contact exclusions. An encounter with the other-colored bank does not create a repeated point of this side curve: that other bank is not part of its boundary traversal. The two side disks may touch each other in their closed exteriors. They remain joined to the back through the two *open* seams, not through the hit point. Figure 1 records the order of these pieces.

This decomposition proves kernel convergence: compact subsets of each side interior, the back, and the open seams persist; compact joining paths in the displayed union also persist. It specifies the limiting prime ends. On the interior of each piece of side boundary one uses its Jordan access, even when another bank touches its image. Near an attaching endpoint one joins the Jordan neighborhoods through the incident open seam. At the hit point the two neighborhoods are joined through the back. Thus the boundary traversal is continuous in exactly the order of the discrete cycles.

For the required uniform convergence of maps, use the admissible domain version of boundary convergence in [9, Lemma A.1 and Corollaries A.1–A.2]. Its hypotheses are as follows: the boundary is the ordered union of three individually simple arcs J_1, J_2, J_3 ; J_3 avoids the interiors of the other two arcs and is accessible from the exterior at each of its points; and the three corresponding boundary arcs converge in the uniform curve metric. The other two arcs are allowed to touch. Here choose J_3 to be a closed arc around the target strictly inside the untouched round part of the boundary. The remaining boundary, divided at the hit point, gives J_1, J_2 . Their separate simplicity and their convergence were just proved; J_3 has exterior accesses in the round disk and misses their interiors. The approximating Jordan graph disks meet the same conditions. A normalization point can be fixed in the back, and the two marked version gives convergence of the start and target prime ends. The cited corollaries therefore give uniform convergence of maps on the closed disk. Composing with the preceding uniformly convergent maps proves the physical

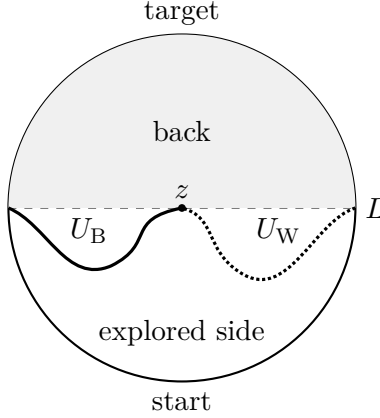


Figure 1: The stopped target domain in current disk coordinates, schematically. The two side domains U_B, U_W attach to the back along the open parts of L on either side of the hit z . Their simple colored boundary arcs may touch in physical projection; the accesses in the displayed boundary order remain distinct. The picture suppresses portions already cut off by the exploration.

statement. Equivalently, marks chosen in boundary order converge because a sequence of boundary subarcs of diameter tending to zero cannot have preimages of positive diameter: that would make a nonconstant limiting parameter arc constant under the limiting map. $\square$

We now identify the law of these domains. The argument also applies when the preceding coordinate map has a nonsimple boundary image, provided its two banks are separately simple and its closed-disk maps converge uniformly. This qualification matters after the first stop.

Lemma 3.7 (Cardy determines the stopped kernel). *The joint limiting law of the hit on L and the two side domains of Lemma 3.6 is uniquely determined by the scalar Cardy limits. For a semicircle cut in coordinates with start 0 and target ∞ , it is the law of chordal SLE_6 stopped when it first reaches that semicircle.*

Proof. First fix a mark in the interior of L . The event of a black connection from the black bank to the separator between that mark and the white end forces the hit to the corresponding side of the mark. The complementary white connection gives the reverse bound. This is the elementary Hex separation in the front disk: the two colored paths along the explored banks terminate at the neighbors of the first separator vertex about to be queried. The white path separates the black bank from separator accesses beyond that vertex toward the white end, and the black path gives the symmetric separation. The comparison is used with a small slack at the mark, so no identity for the individual terminal triangle is needed.

These two comparisons are within the scope of the scalar theorem. Trim each target arc slightly short of its endpoints, and crop a subdisk connecting it to its mate before extending the targets for a lower crossing bound. One target lies on a separately simple bank and the other lies in the interior separator. Their cropped test has Jordan image and stays off the other boundary accesses. The Jordan sandwich of Lemma 2.5 and the conformal-coordinate conclusion of Lemma 2.8 apply to it. In particular this is not an application of Cardy's theorem to an unidentified covering surface. As the cropping and the slack decrease, the two bounds approach the complementary Cardy values. End stops are negligible: the endpoint is on an open monochromatic bank away from the switches, and an approach there carries a free arm of the opposite color out of its neighborhood. The

limiting hit distribution is therefore Cardy's hitting distribution and has no atoms. The same argument works for piecewise smooth indentations of L .

To determine more than the hit marginal, attach finitely many closed Jordan bumps to disjoint subintervals of L on its front side. Their sides are simple piecewise smooth curves, and except for their feet the bumps are interior. Keep a specified interval of possible hits disjoint from all feet. Replace the relevant pieces of L by the bump sides. The preceding hitting argument determines the limiting probability of reaching the retained interval on this indented separator. On this event the exploration has avoided the bumps and the retained hit is also its original hit on L .

The comparison can be made on the same graph and the same bits. Approximate the bump sides by simple graph paths in the front disk, joining the separator in order. These paths agree with the old separator away from shrinking neighborhoods of the junctions. Approach each junction from an interior portion of the bump side and trim at the original crosscut; the endpoint vertices may vary inside those shrinking neighborhoods. There is consequently no requirement to approximate a prescribed smooth arc through a prescribed microscopic vertex. These nested graph separators really hide the bump interiors from the original start side. Before querying a vertex on the new separator the exploration cannot query a hidden part of the old one. On the unchanged retained interval the stopping rules agree.

It follows that the indented Cardy value lies between the limiting probability of strict inclusion of the bumps in the side domains, with hit in the open retained interval, and the probability of inclusion in their closures, with hit in the closed interval. Here *strict inclusion* means inclusion with positive clearance from the side boundary, allowing the foot to continue through its open seam. Conversely strict inclusion and the retained hit force the indented test to pass, by the domain convergence of Lemma 3.6. Enlarge the bumps slightly by regular parallel or polygonal approximations. Inclusion of an enlarged closed bump implies strict inclusion of the original; the Cardy values converge as the enlargement decreases. Since hit endpoints have zero mass, these sandwiches determine the strict inclusion probabilities. The hit's position relative to each foot specifies which side domain contains that bump.

These tests determine both open domains, including their joint law. Indeed a rational ball compactly contained in a side domain can be joined to its open seam by a polygonal finger of positive width whose closure stays in that domain. A countable collection of rational fingers and balls tests all such inclusions. Finite intersections are still bump tests: slightly enlarge overlapping bumps on the same side, fuse them, fill bounded holes, and fill pockets against the intervening seam intervals. Their containing side domain is Jordan, so these fillings are forced by inclusion. Under strict inclusion the enlargement can be taken with positive clearance. Bumps from opposite sides that intersect are incompatible; a filling that covers the retained hit likewise gives an empty event. Subdivide the retained hit intervals to distinguish these cases. Thus the preceding probabilities give a measure-determining class. The simple boundaries then recover the two ordered arcs from their open domains.

Finally compare this characterized law to the classical SLE law in a round disk. The Cardy hitting and hull-avoidance characterization is stated in [9, Section 5, equations (8)–(9), and Lemma 5.1]. The single-semicircle hulls have the side-boundary and first-contact properties just used; these may also be read from the ordinary triangular-lattice coupling in [9, Lemmas 6.3–6.4 and Theorem 5]. Here this last result is used solely to supply properties of the established SLE_6 law in the standard Jordan setting. For that law the indented test is precisely hitting the retained interval before the bumps, so it has the same Cardy value. The measure-determining argument identifies our kernel with that SLE kernel. $\square$

3.3 Iteration and recovery of the ordered path

After a stop, explore in the target face using the boundary colors already supplied by the same band. Conditional on all previous queries, its unknown colors are still independent and fair. There is nevertheless a geometric condition to verify before applying Lemma 3.7 again: the two physical banks of the new domain must remain separately simple. Uniform convergence of the maps alone would not imply this.

View all finitely many stopped bands in the initial Jordan disk coordinates. They constitute the original exploration continued in its target face; no additional sites have been queried in advance. Thus the face-boundary contact rules of Lemma 3.4 still apply. A bulk self-return of one colored arc would force six arms, and a self-return at an original bank is excluded by the corresponding boundary rules. The remaining possibility involves the terminal point of an arc at one of the stops. The hit is in the interior of the preceding crosscut, and hence strictly inside the preceding physical domain. Its map is one-to-one in a neighborhood of that point. The single-step simplicity conclusion therefore excludes such a return as well. At the original target use the fixed-point rule. This proves separate bank simplicity at every fixed finite stage; it does not claim that arbitrary pairs of touching arcs could be glued to make an admissible domain.

The one-step convergence proved above holds along every deterministic converging sequence of inputs satisfying these conditions. It therefore iterates under bounded integration. Explicitly, couple the finitely many preceding domains, marks, and maps to converge almost surely along a subsequence. Conditional on these data, the next colors have the product law on the unqueried sites. For almost every converging input sequence the one-step conditional law converges to the stated SLE kernel. Bounded convergence then gives the next joint law. Crosscuts may be chosen measurably, for example by taking the first graph approximation in a fixed enumeration meeting the prescribed clearance and order conditions.

In upper half-plane coordinates normalize each new map at infinity with derivative one, then translate the new start to 0. The domains in these coordinates agree with the half-plane near infinity throughout a step, so reflection there makes this normalization continuous. No boundary derivative in physical coordinates is used. For each fixed cut radius $v > 0$ and each fixed number of steps, the sampled target domains and tips therefore converge to the corresponding samples of chordal SLE₆, by its domain Markov property.

It remains to show that identifying these samples identifies the whole curve. A small capacity increment alone would not suffice: one must exclude an excursion invisible to the sampled hulls. We give this last argument explicitly.

Lemma 3.8 (No disconnection at a new interior tip). *Let a continuous chordal curve in a Jordan disk generate its filled target hulls. At a finite time before its first visit to the target, at which its tip is interior and has not been visited previously, no point at positive distance from the trace up to that time and from the initial boundary can become disconnected from the target for the first time.*

Proof. Fix such a point w and time t . Take a small open boundary arc around the target whose neighborhood is disjoint from the trace up to time t . Allow connections through that arc to the exterior, and include the complementary closed boundary arc in the obstruction. The obstruction K_t is the image of a continuous curve obtained by traversing that closed arc and then the exploration. Its endpoint, the current tip, is visited only once. Disconnection of w from the target is now separation of w from infinity by K_t .

We need a small local-connectivity observation with the terminal point excluded. For every $\rho > 0$ there is $d > 0$ such that any two points of K_t other than its endpoint at distance less than d can be joined on K_t by a path of diameter less than ρ contained in some strictly earlier obstruction.

Otherwise take counterexample pairs converging to a common point and their curve parameters converging to limiting parameters. If both limiting parameters are terminal, the short terminal subpath joins the pair. If neither is terminal, join each point along its short parameter subpath to the common limiting value and concatenate; both subpaths lie strictly before the terminal time. The mixed case would give an earlier visit to the endpoint, contrary to the assumption. Uniform continuity proves the claimed diameter bound in each case, a contradiction.

If K_t separates w , polygonal closed walks of nonzero winding number about w can be drawn arbitrarily close to K_t . For example take a fine square grid in general position, form the union of closed squares meeting K_t , and trace the boundary of the bounded complementary component containing w . Choose the grid finer than a fixed fraction of $\text{dist}(w, K_t)$. Subdivide such a walk, replace its vertices by nearby points of K_t other than the endpoint, and join consecutive replacement points by the small paths just constructed. The resulting closed walk has the same winding number: each replacement remains in a small neighborhood avoiding w , and can be joined to the original small segment there. It consists of finitely many paths, each lying strictly before time t , and is therefore contained in $K_{t'}$ for some $t' < t$. Nonzero winding implies that $K_{t'}$ already separated w . This contradicts first disconnection at t . $\square$

Lemma 3.9 (Sampled target domains determine the curve). *Every subsequential limit of the exploration in Proposition 3.1 has the law of chordal SLE_6 as an ordered curve.*

Proof. Let γ be such a limit, in the initial disk coordinates. For a fixed sampling radius v , couple its countably many sampled domains with an SLE trace ξ having their law; the latter can be sampled conditionally on the domains. A diagonal extraction over finite numbers of steps gives all the samples in their order on γ . At every sample, the path passes through its tip, the preceding part of γ avoids the open target domain, and the following part lies in its closure. These are direct limits of the corresponding band inclusions, using the domain convergence above.

For fixed v , the SLE capacity increments between samples are independent and identically distributed, strictly positive, and bounded above by Cv^2 . The upper bound is the elementary half-plane capacity bound for a hull contained in a half-disk of radius v ; positivity and independence follow from the nonconstant chordal SLE trace and its domain Markov property with the stated normalization. Their sums therefore diverge almost surely. SLE continuity at the start and transience to its target allow its time interval to be compactified, with the target as the final point.

Take a sequence $v \downarrow 0$. For each v retain the closed monotone correspondence between the positions of the samples on the parameter intervals of γ and ξ , including both endpoint pairs. Take a subsequential Hausdorff limit of these correspondences, using tight parametrizations of the two curves. At the same time retain, at every rational SLE time s , both the compact closure $\overline{D_s}$ of the SLE target domain and its closed complement. Encoding both is necessary to pass the two band inclusions. The joint law of the SLE curve with these closed sets is independent of v , so every subsequential limit retains the target domains of its SLE coordinate. No continuity of the map from traces to target domains is needed. The sample capacity gaps are at most Cv^2 ; hence the limiting correspondence has full projection on the SLE parameter interval. Its paired points coincide in space. Tail sample pairs accumulate at the target, so the final endpoint causes no missing projection.

SLE has no constant time interval. A value of the γ parameter cannot therefore be paired with two distinct SLE times. Any interval missing from the first projection has its two endpoints paired with the same SLE time t . If the correspondence does not already identify the two curves up to pauses, there is consequently a nonconstant excursion of γ that leaves and returns to $\xi(t)$

inside such a missing interval. Every point w on this excursion satisfies

$$w \in \bigcap_{s < t} \overline{D_s}, \quad w \notin \bigcup_{s > t} D_s. \quad (19)$$

It suffices here to use rational s . For the first inclusion, apply the suffix condition at a sample between s and t ; for the second, use the prefix condition at a sample between t and s . Monotonicity of the target domains and the retained closed sets pass these conditions to the limit.

If $\xi(t)$ is on the initial boundary, the excursion violates the absence of a boundary double visit of γ . If it is interior and also visited by SLE at another time, the full projection of the correspondence supplies a third visit of γ in addition to the two excursion endpoints. This violates the absence of interior triple points. We are left with a uniquely visited interior tip.

A point satisfying (19) that is off the SLE trace and off the initial boundary would have disconnection time exactly t . Indeed it has a neighborhood free of trace; while a connection to the target exists, a compact connecting path remains open for a short later time. The two inclusions say respectively that disconnection has not happened earlier and that it has happened immediately afterward. Lemma 3.8 excludes this possibility at a uniquely visited interior tip. The hidden excursion must therefore lie entirely on the SLE trace or the initial boundary.

Take a small interior neighborhood of $\xi(t)$. Every other point of the hidden excursion there that lies on the SLE trace has another visit supplied by the correspondence, and is a double point of γ . A nonconstant excursion leaving $\xi(t)$ must cross one of a fixed countable collection of line segments forming arbitrarily fine rational grids inside this neighborhood. That crossing would be an interior double point on a fixed segment, excluded by Lemma 3.4. Thus the excursion is constant. No missing nonconstant interval remains, and the ordered curve is SLE. $\square$

Lemmas 3.3 and 3.9 prove Proposition 3.1. Notice that the proof gives convergence along every deterministic sequence of Jordan graph domains with ordered boundary convergence, which is the form needed to explore the random faces below.

3.4 Restoring a virtual bank and exploring loops

We now work in a large Jordan disk, with all eventual crossing tests a positive distance from its boundary. Initially give its boundary a single color. The disk may be approximated by internally chord-free site cycles. Changing boundary colors on that cycle does not change the original fair colors of sites in the region of the tests.

The continuum object used here is the oriented percolation-loop construction of Camia and Newman [8]: successive chordal SLE₆ explorations in Jordan domains, with the two sides of each interface assigned their colors. We recall the discrete construction sufficiently precisely to verify that it has this limit.

With actual two-switch boundary conditions, explore and retain the monochromatic unknown faces of Lemma 3.2. Inside a face with constant boundary color, temporarily change one boundary bank to the opposite color and make a chordal exploration. We call that changed bank *virtual*, because its original color is restored afterward. For definiteness suppose its virtual color is black and its actual color white. Excursions away from the virtual bank supply pieces of actual interfaces; further explorations in the faces incident with that bank complete those pieces to loops. The following incidence description specifies the completion, including faces attached at only one site.

Lemma 3.10 (Restoration and ordered excursions). *Restoring a virtual boundary bank pairs its excursions with explorations of the incident unknown faces according to their order along the bank.*

For every fixed positive diameter cutoff, the excursions, the incident faces used to complete them, and their completed oriented loops converge jointly. No macroscopic loop is produced in the limit solely by excursions or incident faces of vanishing size at this step. Conditional on the exposed boundaries, all still unknown interior colors remain independent and fair.

Proof. First work at a fixed mesh, with a virtual black bank whose actual color is white. List the edges from that bank to revealed vertices off the bank, including the switch edges. Order them along the bank and then within successive interior vertex stars. Thus several entries may use one bank vertex. Label each entry by the color of its *off-bank endpoint*. The white entries are exactly the places where the virtual wall crosses a bank-incident edge. Their order along the wall is the bank/star order: the short spokes joining them to the bank lie on one side of the simple wall and cannot cross.

Between two consecutive white entries the list has the form

$$W_i, B_1, \dots, B_k, W_{i+1}.$$

Write E_i for the virtual excursion between those white entries and I_i for the intervening bank interval, retaining the star order if the interval is a single vertex. Apart from its end triangles, E_i is an actual interface after restoration. We describe the returning chain that completes it.

Consecutive entries bound a common face, possibly a triangle, because there are no rim chords and every incident face has a single bank interval or vertex by Lemma 3.2. For $k > 0$, each middle wedge B_j, B_{j+1} is a triangle or an unknown face whose boundary is black before restoration. Restoring the bank to white creates exactly two switches there. Join them directly through the triangle, or by exploring that unknown face with its actual two-switch boundary conditions. The end wedges W_i, B_1 and B_k, W_{i+1} have mixed colors before restoration, so they cannot be unknown monochromatic faces; they are band triangles. After restoration they join the returning chain to the surviving ends of E_i . Consequently a nonempty run completes that same excursion to one real loop. An empty run through only bank triangles has no macroscopic contribution. White entries separate different runs, so this operation cannot concatenate distinct excursions. Unused interfaces lie in the other unknown faces, including residual faces of these closing explorations. This is a finite incidence identity; it asserts no independence of the virtual excursions.

We next show that the identity passes to the limit. Work in the initial disk coordinates of this step. A sequence of excursions of diameter bounded below has, by curve tightness, a limiting pair of endpoints and an ordered limiting arc. Its interior cannot acquire a contact with the virtual bank: during the discrete excursion neither bank shadow uses that bank, so such a contact between positive excursions gives three free half-plane arms as in Lemma 3.4. Conversely a compact subarc that avoids the bank in the limit eventually avoids it discretely. There is no boundary interval of the limiting wall, no boundary double visit, and no return at a switch. Hence positive-size excursions have distinct endpoints and match in their order; no positive excursion can collapse to a spike at one boundary point.

Every limiting contact with the open bank is approximated by an actual recorded occurrence, not just the contacts selected by a diameter cutoff. Otherwise the approaching and departing wall segments in an occurrence-free neighborhood would give the same three-arm event. Together with the initial and final bank points, this recovers the full order of the contacts.

For a matched excursion, the intervening bank interval is incident with one limiting face of positive size. Every compact subinterval of that interval has an adjacent strip avoiding the band; its limiting face has a Jordan boundary by Lemma 3.5. Its two attachments converge to the excursion endpoints, so its actual two-switch marks also converge. Proposition 3.1 supplies the

closing exploration. Any other incident faces involved in that run have attachments collapsing to the endpoint neighborhoods. Their diameters tend to zero: a positive diameter face of the virtual bank's color with a collapsing attachment interval, including a singleton, is excluded by Lemma 3.4. Thus their pieces cause no error in the curve metric when the loop is pasted.

The same argument rules out a surplus macroscopic loop completed from vanishing excursions at this step. If E_i shrinks, its endpoints coalesce, and continuity of the inverse Jordan-bank parametrization makes I_i shrink as well. Every completion face for this run is attached to I_i ; all its band triangles are within a vanishing mesh distance of that interval. The diameter of the completed loop is therefore bounded by the diameter of E_i , the diameter of I_i , twice the largest completion-face diameter, and a vanishing mesh error. The collapsing-attachment exclusion makes that largest diameter tend to zero. This also excludes an accumulation of many small completion pieces into a macroscopic loop. An other-colored face whose boundary merely touches the bank in the limit, without genuine cycle edges on it, receives no new switch assignment. True intervals of attachment have been distinguished by Lemma 3.5. This distinction prevents an incorrect extra reconnection at a limiting contact.

Actual two-switch explorations leave monochromatic faces again. Their unknown sites are fresh conditional on all queried colors by the product-law argument preceding Lemma 3.2. Select and match faces, for example, using interior points from a fixed dense countable set that avoid the traces. Along every deterministic converging sequence of selected Jordan faces, Proposition 3.1 applies. The bounded-integration argument used for stopped kernels therefore gives joint convergence of each finite set of operations. Uniform continuity of the Jordan coordinate maps transfers all these statements back to physical coordinates. $\square$

Proposition 3.11 (Oriented loop convergence). *In a Jordan disk with monochromatic boundary, along every deterministic sequence of tilings under consideration, the critical interface loops converge to the Camia–Newman SLE_6 loop construction. For each positive cutoff chosen outside the atoms of limiting loop diameters, the loops above that cutoff can be matched in the curve metric, including their orientation, multiplicity, and order of excursions. There are no surplus macroscopic loops. The convergence holds jointly for any finite number of independent color replicas on the same fixed geometry.*

Proof. Lemma 3.10, iterated, proves convergence of every fixed finite number of recursive operations. It remains to justify that finitely many operations discover all loops above a fixed scale with arbitrarily high probability. We use a property of the already identified continuum SLE construction, checking its precise scope.

Fix a conformal map from the unit disk onto the initial Jordan domain, extended as a homeomorphism of closures. In these initial disk coordinates choose the exploration order and boundary endpoints as in [8, Section 3.2]: priority is by maximal coordinate span, with its specified dense-point tie breaking; monochromatic domains use the maximal-span endpoint rule, and two-color excursion domains use their two prescribed switches. Start with that construction's monochromatic disk and virtual first exploration and transport the recursion by the fixed map. For this continuum recursion the number of steps until every remaining domain has coordinate diameter below a fixed $r > 0$ is almost surely finite. Here is the exact consequence of [8] that proves this assertion. Theorem 5 there gives finite-step coupling with the reorganized triangular construction, and its proof, equation (20), proves tightness of the stronger triangular stopping count until all remaining domains have coordinate span below a prescribed cutoff. The final part of that proof transfers this estimate from the original construction to the reorganized one by its maximal-span comparison.

To pass that statement to the continuum, suppose that after n steps a continuum remaining domain has diameter greater than r . Choose a compact connected subset of its interior with diameter greater than $r - z$, by joining two interior points with a compact path. This subset has positive distance from the finitely many explored boundaries. Under the finite-step coupling it stays in a corresponding triangular remaining domain for all sufficiently small lattice meshes. Hence that triangular domain has diameter greater than $r - z$, and its coordinate span exceeds $(r - z)/\sqrt{2}$. Taking the mesh limit in the tight stopping-count estimate, and then $n \rightarrow \infty$, proves the continuum assertion, using a smaller cutoff to obtain the strict diameter inequality. This use of the triangular theorem establishes a fact about the known continuum SLE recursion; it supplies no convergence assertion for the present tessellations.

Uniform continuity of the fixed initial conformal map converts this coordinate exhaustion into exhaustion at every positive physical diameter. We may now couple finitely many steps of our construction to those continuum choices. On each matched boundary choose the nearest ordered discrete endpoints. The choice depends only on the coupled past, so it leaves the unknown discrete colors with their correct conditional product law. This avoids imposing an unwarranted continuity assumption on a maximizing endpoint rule. At each step Lemma 3.5 matches the remaining macroscopic faces with no surplus. Induction gives that property after every fixed finite number of operations.

Fix an error probability and first choose, using continuum exhaustion, a deterministic number of operations after which all continuum remaining faces are smaller than the desired cutoff with the stated high probability. If an excursion already drawn still needs its matching face exploration to complete a macroscopic loop, perform those additional operations as well. For each positive excursion cutoff there are only finitely many such mates; include all of them and then decrease this auxiliary cutoff. The no-surplus assertion of Lemma 3.10 ensures that vanishing excursions at any of these finitely many steps cannot produce a missing macroscopic loop. All other loops lie within the remaining small faces. Finite-step convergence now gives the required loop matching, and then the error probability and auxiliary cutoffs may decrease. The incidence rule retains orientation and multiplicity throughout. Finally, on deterministic geometry different replicas have independent colors, so their finite recursive laws are independent products; the same limiting and cutoff arguments give their joint convergence. $\square$

We have obtained a common critical loop law from the scalar crossing limit, with the fixed geometry retained throughout each exploration. To use it in the near-critical expansion, we next read the prescribed polygon crossings from these loops and allow finitely many fixed squares to be forced to specified colors.

4 Joint crossings and prescribed regions

Section 3 identifies the oriented critical loops along each deterministic good sequence. We now prove convergence of the observables needed for the near-critical argument: finite collections of polygon crossings after several separated squares have been forced black or white. First we recover the literal crossings confined to each polygon from the oriented loops. We then combine continuity under small motions of the prescribed regions with finite witnesses made of ordinary polygon crossings. Proposition 4.5 collects the resulting common joint law, including its conditional version for Poisson geometry.

4.1 Reading joint crossing tests from the loops

An interface may enter and leave a quad repeatedly. We therefore determine the quad crossing from the boundary pairing of interface portions inside it.

Write the sides of a quad in boundary order as S_1, S_2, S_3, S_4 , with S_1, S_3 the black target sides. At a finite mesh, draw the interfaces along cell edges, retaining their orientations and hence the color on each bank. In general position with respect to a polygon, their portions in its interior are disjoint crosscuts and closed curves. The following elementary rule reads the crossing from these portions.

Lemma 4.1 (Boundary-pairing rule). *Suppose finitely many disjoint interfaces, with their two bank colors, cut a Jordan quad, avoid its marks, and cross its boundary whenever they meet it. If an interface crosscut joins S_1 to S_3 , the black bank supplies a black crossing. If one joins S_2 to S_4 , the white bank supplies a complementary white crossing. If neither type occurs, there is a complementary region incident to all four sides; its color determines the crossing. Whenever a crosscut joins the two sides incident to a mark, the orientation of the outermost such corner cut determines this central color.*

Proof. A crosscut between opposite sides separates the other pair. Paths running arbitrarily close to its banks give the asserted color connections; they can be taken with clearance on their respective sides. The two types of opposite crosscut cannot coexist, because they would intersect.

In their absence every crosscut has both ends on one side or on consecutive sides. On the boundary circle the endpoint pairing is noncrossing. A same-side pair cuts off a pocket containing no mark. A pair on consecutive sides cuts off the corresponding corner. Remove the pockets and, for each corner that is cut off, its outermost corner pocket. The remaining region is connected and incident to each of the four side intervals between the removed pockets. This can also be proved by induction: remove an innermost paired interval first, then restore it as a pocket along one boundary interval of the remaining region. Interior closed curves only remove interior pockets. Consequently the region left between the corner pockets meets all four sides. A path through it joins either pair of sides, and its color decides which crossing occurs. The bank of an outermost corner cut facing this region has that color. All paths used here lie in the quad; no identification of different boundary points or external connection is made. $\square$

We explain why this finite rule is stable for the limiting loops. First put the polygon in general position by a translation chosen from an arbitrarily small open disk. For almost every such translation, almost surely the limit loops avoid the vertices and marks, have no double point or mutual contact on a side, and have no tangency there that is a local extremum of the coordinate normal to the side. Here are the estimates and deterministic facts behind these assertions. A prescribed point is avoided by the one-arm estimate. A double visit or a contact of two loops on a line segment produces four traversals in a ball centered on that segment. Covering the segment by $O(r^{-1})$ balls and using Lemma 2.7 gives an upper bound $O(r^c)$, for some $c > 0$, when the traversals continue to a fixed positive distance. Let that distance run through a countable sequence and then let $r \downarrow 0$. Finally, for any continuous real function, the set of its local-maximum and local-minimum values is countable: each such value is the maximum or minimum on some interval with rational endpoints. Apply this to the normal coordinate along each of the countably many loops. Almost every translation avoids all those levels. These facts hold simultaneously for a finite collection of polygons. General position at finite meshes is obtained by excluding a further null set of translations; at a Voronoi vertex the incident cells are pairwise side-adjacent, so this exclusion introduces no new connectivity convention.

There are also corner cuts at positive, possibly random, distances from each mark. To see this, choose many disjoint buffered annuli around the mark, all small enough that the two incident polygon sides are the only boundary pieces present. In each annulus, fixed crossing chains give a black circuit and a white circuit in separated subannuli with a probability bounded below. The colors in these buffered regions are independent on a deterministic mesh, once the mesh is sufficiently small. Thus the probability that none of k annuli supplies the circuits is at most $(1 - c)^k$. The two circuits force an intervening closed wall enclosing the mark. Following this wall through the polygonal sector gives a cut between the incident sides. Take the mesh limit first and then $k \rightarrow \infty$. This proves the claim also when the mark lies in the interior of a straight side.

Fix one such cut at each mark, and trim its two reaches on the incident sides a positive distance away from that mark. Any outermost corner cut beyond these reaches has positive diameter: the two compact side portions in question are disjoint. Opposite-side excursions also have positive diameter after the mark neighborhoods have been removed. At any fixed positive reach there are only finitely many relevant excursions, by curve continuity and the finite macroscopic matching in Proposition 3.11. Their orders and bank colors persist under convergence. Indeed their compact interior portions stay in the open quad. At an endpoint on a side, the loop must approach the line from both sides in every sufficiently small parameter neighborhood, unless that line is a local-extremum level. Therefore nearby curves recover the endpoint crossing. Conversely, a sequence of excursions of positive reach has a limiting excursion: it cannot collapse at a mark, and it cannot acquire an intermediate boundary contact without either a double contact on the side or a forbidden one-sided tangency. Reach cutoffs can be chosen to avoid equality. The boundary-pairing rule now gives the same crossing decision on matched configurations. Unmatched loops of vanishing diameter cannot change it, because the positive corner reaches have already fixed the central color in the case with no opposite-side excursion.

For a specified, unperturbed polygon use the inner and outer Jordan sandwich tests from Lemma 2.5. They are chosen so that, on the same colors, the lower crossing implies the specified crossing, which implies the upper crossing. The tests may be put in the preceding general position. The difference between their limiting probabilities tends to zero with the sandwich width, by continuity of Cardy's formula under convergence of the marked Jordan boundaries. A union bound applies to any finite family. We have therefore proved joint convergence of the exact closed-cell crossing indicators, along every deterministic good sequence, to a common critical law. This argument also proves convergence for polygonal site-path tests with the terminal conventions of Section 3: the positive sandwich margins absorb the conversion between site paths and cell paths.

4.2 Continuity under changes of prescribed regions

Fix a finite family of closed squares

$$H_j = \{x \in \mathbb{R}^2 : \|x - z_j\|_\infty \leq h_j\}, \quad h_j > 0, \quad 1 \leq j \leq J.$$

Assume that the squares are mutually disjoint at positive distance and that each is at positive distance from every polygon boundary under consideration. For each crossing test choose, separately, whether to force all cells meeting H_j black, to force them white, or to leave them unchanged. A square outside a particular quad is irrelevant once the mesh is small. Different tests may make different choices while using the same original fair colors. If a cell meets two differently prescribed squares, assign it the color of the square with smaller index; this convention is eventually irrelevant on a good sequence, since the squares have positive separation. Throughout this subsection the number of squares, their positive radii, and their separations are fixed before the mesh tends to zero.

Call an arm *free* when it uses only sites whose original fair colors have not been prescribed in the test. We will use the mesh-first bounds

$$\begin{aligned}\limsup_{e \rightarrow 0} \mathbb{P}(\text{one free arm from } r \text{ to } R) &\leq C(r/R)^\gamma, \\ \limsup_{e \rightarrow 0} \mathbb{P}(\text{three free half-plane arms}) &\leq C(r/R)^\beta,\end{aligned}\tag{20}$$

where $\gamma > 0$ and $\beta > 1$; the second bound allows any color order and any fixed side direction. These are consequences of Lemmas 2.1 and 2.11. Shifts of the bounding line and constant factors in the radii are allowed. All probabilities in the next two lemmas are color probabilities on a deterministic good sequence.

Lemma 4.2 (Continuity of prescribed-region tests). *For any fixed finite family of polygon crossing tests with the square prescriptions above, the probability that a test changes under a sufficiently small polygonal displacement of its sides, marks, or prescribed squares tends to zero, uniformly in the mesh-first limit. The displacements may change the centers and positive side lengths of the squares, provided the stated separations persist. The same conclusion holds simultaneously for any fixed finite collection of prescription choices on the same colors.*

Proof. We give both the geometric interpolation and the arm test for a single local change. Choose $u > 0$ smaller than the separations of all unrelated sides and squares, and put balls of radius Cu around the polygon vertices, marks, and square corners. Outside these balls divide each straight edge into $O(s^{-1})$ intervals of length comparable to s , where $s \ll u$. For a displacement much smaller than s , perform the motion one interval at a time, using a piecewise linear cutoff supported on a slightly longer interval. Leave the portions within distance $2u$ of the vertices for the corner moves, and taper the edge moves between distances $2u$ and $3u$; motions in balls of radius $4u$ then complete the interpolation. There are boundedly many corner moves and $O(s^{-1})$ edge moves. A polygonal collar identifies the vicinity of each edge with a straight strip. Interpolating its sufficiently small piecewise linear ambient displacements preserves the Jordan property; it also gives the familiar quad sandwiches by shifting target and non-target sides in their respective directions. Squares are treated by the same construction. Each intermediate prescribed region remains a connected disk, and the part common to the two regions in a local comparison is connected. This last property can be ensured by taking the motions to be graphs in their collars and retaining the unchanged inner square. All constructions are fixed before $e \rightarrow 0$.

Consider an edge move supported in a ball of radius Cs . If it changes a crossing, take a black crossing from the configuration where the crossing occurs and an open white complementary crossing from the other configuration. Outside the changed neighborhood their paths run in the common tiling and coloring. For a quad-side move, their portions going to the other three sides give three disjoint free arms from scale Cs to scale cu . They lie in a half-plane translated by $O(s)$ from the original supporting line. Trimming at first and last hits of auxiliary curves preserves the order of the arms. The paths stay in the common region of the two domains; using the enlarged half-plane therefore does not require the intermediate boundary itself to be straight.

At an edge of a black prescribed square, the two arms of the white complementary path are free. At least one of the two black branches towards the target sides is also free until scale cu . Otherwise both target sides would connect, before entering the changed neighborhood, to the common black prescribed region. That region is connected also in the comparison without a crossing, so it would already join the target sides there, a contradiction. The white branches cannot connect to one another in the common part so as to avoid the changed region, since they would then block the black crossing in the other comparison. After proper trimming we thus have three disjoint free

arms in the exterior half-plane of the square edge. A white prescribed square is handled with the colors interchanged. The open white connections through a white prescribed region exist with clearance, because its forced cells form a connected union by side adjacency.

Here and below the “cells meeting” convention is compatible with that connectivity. Follow a path inside the connected prescribed region through the cells it meets. Consecutive cells share a side, or meet at a tiling vertex where the three cells are pairwise side-adjacent. The resulting side-adjacency chain connects all cells meeting the region. A path in those cells can be moved into cell interiors and relative interiors of shared sides. On an unchanged straight boundary piece, convexity of each cell and the shortcut argument for crossing and complementary crossing remove possible reuse of a cell by two candidate arms. The finitely many bends introduced by a fixed interpolation require one further observation. A shared cell there, despite the arms continuing to a fixed outer scale, would give a free arm from a vanishing neighborhood of that bend to that scale. Its probability tends to zero by the one-arm bound. Thus the graph-arm extraction is valid with vanishing error, including for the exact closed-cell crossing convention.

At a corner move, a changed crossing requires at least one free arm from scale Cu to a fixed positive scale. At a quad vertex or mark this is a branch of the crossing towards a target arc away from that vertex. At a prescribed square it is a branch of the color opposite to that square; this branch cannot use the prescribed region. Since all other prescribed regions and unrelated boundary pieces are at fixed positive distances, the arm is free throughout the annulus used for this estimate.

A union bound and (20) give, for displacement small enough in terms of u, s ,

$$\limsup_{e \rightarrow 0} \mathbb{P}(\text{some compared indicator differs}) \leq Cu^\gamma + Cs^{-1}(s/u)^\beta. \quad (21)$$

Constants may depend on the fixed tests and their separations. The vanishing errors at the finitely many interpolation vertices have already disappeared in the mesh limit. Given an error tolerance, first choose u so that the corner term is small, then choose s so that the edge term is small, and finally restrict the displacement. Equivalently, in the displayed bound the limits are taken as $e \rightarrow 0$, then $s \downarrow 0$ at fixed u , and then $u \downarrow 0$. This proves the lemma. $\square$

4.3 Finite witnesses after prescribing square colors

Continuity alone does not identify the law of the modified crossings: prescribing a square does not make its neighboring crossings independent. We instead show that a positive crossing, after a small favorable perturbation, has a witness consisting of finitely many ordinary critical polygon crossings. Those were identified in Subsection 4.1.

For a black crossing, a *favorable perturbation* enlarges every black prescribed square, shrinks every white prescribed square, and makes a quad sandwich favorable to black. The latter moves its target caps slightly towards each other and its non-target sides outwards in polygonal collar coordinates. A path crossing the original quad then crosses the perturbed quad after it is clipped at the new caps. We use two nested perturbations, called the middle and final perturbations, with positive slack between each pair. All prescribed squares remain separated and off the quad boundary.

Lemma 4.3 (Free segments of a terminal path). *Discard squares left unchanged in the test and squares outside its quad. Construct a graph approximation to the middle quad with the remaining middle squares removed. Give its vertices the original fair colors. Fill each square hole by a fan from one new vertex having that square’s prescribed color, and attach uncolored terminals to the*

two target arcs. On the event that the original modified configuration crosses, this graph has a black terminal path for all sufficiently small meshes. It has such a path whose portions between consecutive terminals or fan vertices form at most $J + 1$ free paths. Their joint laws, restricted to the crossing event, are tight in the curve metric. Every subsequential limiting free path is simple after constant parameter intervals are removed.

Proof. Approximate the outer boundary and hole boundaries in disjoint thin strips by simple site cycles, in boundary order, using the graph approximation construction of Section 2. Hole rims are free vertices. The graph outside these cycles, with each hole triangulated by its fan, is a finite planar triangulation. The terminal neighbors are exactly the sites on the corresponding target arc; the terminals create no connection between other boundary vertices.

Shadow an original closed-cell crossing by a site path with vanishing error. Before it enters a black forced square, it crosses the rim of the enlarged middle square through free black sites; crop there and replace its subsequent passage through that square by the black fan. An original black path stays away from every shrunken white hole. The quad collar slack allows the shadow to be clipped at the middle target arcs and keeps it clear of the middle non-target sides. This produces the claimed terminal path. These statements only use the small mesh of cells on the fixed compact region and the strictly positive slack, so hold for all sufficiently fine configurations on the given crossing event.

Choose an outermost black terminal path on one side. More precisely, add an external edge joining the terminals, take the union of all simple black terminal paths together with that edge, and take the appropriate exterior boundary cycle of this union. Each constituent path with the external edge is a cycle, and these cycles share that edge; their union is two-connected. Its exterior boundary therefore supplies a simple terminal path P . No black path can leave P into the exterior side and return to a distinct vertex of P : taking its first departure and first return would produce a terminal path farther on that side, contrary to the boundary construction. We call this the no-bypass property. It is the same elementary outermost-path construction used in Lemma 2.6.

A simple P visits each fan vertex at most once and never visits a white fan vertex. Remove the fan edges and terminal edges. The remaining segments join distinct rim or terminal sets and consist entirely of free black sites. There are at most $J + 1$ of them. Many traversals of a fixed annulus by one segment yield equally many disjoint free black arms, after proper trimming. The one-color disjoint-occurrence bound and the one-arm bound control these numbers. The successive-excursion compactness argument in Lemma 3.3 consequently applies to these segments, also at their endpoints. Notice that the segments may be selected using all the colors; the bound is on the existence of the disjoint arms and does not assume freshness conditional on the selected path.

We give the additional argument that rules out a self-touch in a limit. Suppose first that the contact lies in the bulk, at positive distance from all rims. Two visits with a nonconstant intervening excursion produce four proper traversals of a sufficiently small annulus, with a fixed positive outer radius. Between the traversals, on the side forbidden by the no-bypass property, there are at least two disjoint radial gaps. Indeed, in cyclic order the four directed traversals have distinct sectors on their forbidden sides; at most two traversals can border the same such sector. Each selected gap contains a white arm. To justify the last assertion without assuming smooth graph cuts, choose two intermediate graph cycles, with wide radial margins. Within the gap each cycle has a transverse vertex path between its two path banks. One way to see this is to label just the vertices of that cycle black and apply Hex in the gap: an avoiding radial white path would cross the separating cycle, which is impossible. Use an innermost such cut at the inner cycle, and then one at the outer cycle. They enclose a quadrilateral whose side banks are portions of P . A

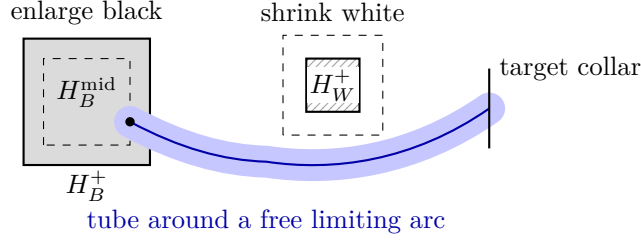


Figure 2: The middle rims are dashed and the final prescribed squares have solid boundaries. A limiting free segment starts on the middle black rim, strictly inside the final black square. Its tube stays clear of the final white square. Its other endpoint is shown in a target collar. Thus an ordinary crossing of a polygon quad in the tube can be joined through the final black region and clipped at the target cap. The figure is schematic; no regularity of the limiting arc beyond simplicity is used.

black crossing between these banks would leave into the forbidden side and return to P ; trim at its first return to obtain a bypass. Hex therefore gives a white crossing between the transverse cuts. With the cuts chosen away from the trimmed ends, the departure points on P are internal and both adjacent path edges are still present, as required for the no-bypass argument. Trim the white crossing to the smaller annulus. Doing this in the two disjoint gaps yields four black and two white site-disjoint arms. All vertices involved are free; no fan edge or rim has entered this annulus.

By Lemma 2.7, the probability of such a six-arm event in any of $O(r^{-2})$ balls of radius r is $O(r^c)$, when its outer radius is fixed. It tends to zero. At a point of a straight rim or outer boundary, a self-touch gives at least three free traversals in the corresponding half-plane. There are three even if one of the two visits is an endpoint of the segment. Covering that side by $O(r^{-1})$ balls and applying Lemma 2.11 again gives a vanishing bound. At the finitely many polygon corners, a single free arm reaching from vanishing scale to a fixed scale already has probability tending to zero. The endpoints of a segment lie in distinct, positively separated rim or terminal sets, so they cannot coincide with each other. These exclusions at countably many outer scales rule out every nontrivial repeated point of a limiting segment. $\square$

The need for a middle perturbation is now visible; see Figure 2. A free segment can touch a middle rim or boundary. The final perturbation puts its black attachment points inside an open black square and separates it from every final white square. Thus the segment can be surrounded by a genuine open tube without having to reconstruct any limiting connectivity at a contact point.

Lemma 4.4 (Transfer by polygon witnesses). *Let two deterministic good sequences be coupled so that their ordinary critical polygon crossing indicators converge jointly to the same indicators for a countable family containing the rational polygons. For any fixed prescribed-square test, the probability of a crossing in the first sequence and no crossing in the final favorable perturbation of the second tends to zero. The conclusion holds for finitely many tests and for independent color replicas.*

Proof. It suffices to consider a subsequence on which the asserted mismatch has positive limiting probability. Include its indicator, the crossing vectors, and the at most $J + 1$ segments of Lemma 4.3 in a joint subsequential representation. Pad the segment list when necessary and include its finite combinatorial itinerary through the hubs as a coordinate in a finite discrete space. In the representation this itinerary stabilizes almost surely, though its limit may be random.

Tightness gives convergence of the segment curves; Lemma 4.3 makes each nonempty limit a simple arc. On the represented mismatch event their ordinary critical crossing vectors agree in the limit, coordinate by coordinate.

For each limiting segment, choose its two end neighborhoods as follows. An end at a black hub lies strictly inside the final enlarged black square; an end at an outer terminal lies in the collar on the appropriate side of the final target cap. The segment stays a positive distance from the final white squares, because these are strictly smaller than the middle white holes. Its non-target boundary approaches have room inside the final non-target collar. Near the two ends, cut the simple arc at points still within the prescribed end neighborhoods. The planar Schoenflies theorem gives a rectangular neighborhood of the intervening simple arc, which can be chosen as a thin tube around it. Choose a polygon quad in this tube whose target end cuts are inside those neighborhoods and whose lateral sides leave room around the arc. A sufficiently close rational polygon approximation preserves these properties. Uniform convergence in curve order implies that the corresponding free segment in the first model crosses this fixed quad for all sufficiently large indices.

The witnesses can depend on the limiting configuration, but are selected from a countable list. On the representation, convergence of a Boolean coordinate means eventual equality of that coordinate. Thus, simultaneously for every member of the list, a witnessed black crossing in the first sequence is also present in the second for all sufficiently large indices. There are only finitely many witnesses on the event under consideration. Their crossings in the second sequence avoid its final white squares: the tube clearance eventually exceeds every cell diameter in the fixed region, so none of the cells in a tube can be recolored white by a distant square. If a crossing enters an enlarged black square, that only helps it. At a hub, the two adjoining witnesses have endpoints in the open square, so the forced black cells connect them there with clearance. At the outer ends, join in their prescribed order and clip the resulting path at the final target caps. This yields a black crossing of the final test, contradicting the represented mismatch. The tubes need not be disjoint: their only required joins occur inside their specified black squares, and any additional intersection of black witnesses is harmless.

For several tests apply this argument to each positive indicator; for an indicator equal to zero use the same implication in the reverse coupling when deriving equality of laws. For replicas keep a separate segment list and crossing vector in each replica. The arm bounds apply separately, and a finite union bound proves the assertion without any assumption that the replicas have independent geometries. $\square$

4.4 The resulting critical comparison

We collect the precise conclusion needed below. A test specification consists of a polygon quad, a replica label, and a choice in $\{*, B, W\}$ for each of a finite family of squares; $*$ means that the original fair colors are left unchanged. A finite collection of specifications may repeat the same quad with different prescriptions. Within a replica all tests use the same fair color configuration.

Proposition 4.5 (Joint critical tests with prescribed squares). *Fix finitely many polygon quads and finitely many closed squares of positive side length, with the squares mutually separated and separated from every quad boundary. For any finite collection of test specifications, the joint law of their exact closed-cell crossing indicators has a common limit along every deterministic good sequence of either model. The limit is continuous in the centers and positive side lengths of the squares throughout the open range where the stated separations hold, and under sufficiently small*

marked polygonal deformations of the quads. The conclusion remains valid when the centers and side lengths converge to such an admissible fixed configuration as the mesh tends to zero.

For Poisson–Voronoi percolation the conditional law given η_e converges in probability, in total variation on this finite Boolean space, to the same deterministic limiting law. With any fixed number of conditionally independent fair-color replicas on one Poisson geometry, the limiting law is that of independent copies of the same critical configuration, with all the prescribed tests performed in their specified replicas. In particular, two replicas sharing their geometry have the same joint critical limit as two independent triangular replicas.

Proof. For unmodified tests the assertion on a deterministic good sequence was proved in Subsection 4.1. On any subsequence, compactness of the countable Boolean crossing space permits a representation in which these ordinary test indicators converge coordinatewise. Couple two candidate subsequential limits through their common unmodified crossing vector. Lemma 4.4 transfers every positive prescribed test from either sequence to the other’s arbitrarily small favorable perturbation. By Lemma 4.2, replacing that perturbation by the specified test costs a probability tending to zero with its size. For precision, let $X_e^{a,i}$ be test i in sequence $a \in \{1, 2\}$, and let $X_e^{a,i,+v}$ denote a favorable perturbation of size $v > 0$. The transfer implication in both directions gives

$$\limsup_{e \rightarrow 0} \mathbb{P}(X_e^1 \neq X_e^2) \leq \sum_i \sum_{a=1}^2 \limsup_{e \rightarrow 0} \mathbb{P}(X_e^{a,i,+v} \neq X_e^{a,i}).$$

The right side tends to zero as $v \downarrow 0$ by continuity. Thus the resulting Boolean vectors agree with probability tending to one under the coupling, so their subsequential laws coincide. This proves existence and commonality of the limit, including different prescriptions in different tests on the same colors.

The continuity assertion follows directly from Lemma 4.2. For mesh-dependent centers or radii, bracket each moving square between two fixed concentric squares with radii arbitrarily close to its limiting radius. Force the inner or outer square according to the monotone direction of its color: enlarging black helps a black crossing, while enlarging white hinders it. With a corresponding quad sandwich if needed, the specified moving test lies between two fixed tests. Continuity makes their disagreement probability vanish. This proves the moving-parameter statement without requiring estimates at mesh-dependent positive scales.

On a deterministic tiling independent color replicas have product laws. Apply the preceding convergence to the finite vector in each replica; their joint law therefore converges to the corresponding product. For Poisson geometry, Lemma 2.5 says that from every vanishing-mesh sequence one can extract a subsequence along which the environments are almost surely good deterministic sequences. On this subsequence, each conditional atom probability for our finite Boolean vector converges almost surely to its just identified deterministic value. There are finitely many atoms, so the conditional total-variation distance tends to zero almost surely. The subsequence criterion proves convergence in probability along the original sequence. The argument applies to the product color measure on the same fixed environment and hence retains the shared geometry in the replica assertion. Averaging gives the corresponding annealed limits. $\square$

5 An absolutely convergent expansion in sites

We now express finite near-critical crossing tests in terms of critical color laws. The expansion is in the sites at which a color history is changed. Its coefficients can have either sign, and two independent color replicas may use the same Poisson geometry. The purpose of this section is to

obtain an absolutely summable bound for all the coefficients and uniform integrability for each fixed coefficient. Identification of their limits will then be a separate local problem.

Recall the fixed rectangle Q_0 and the normalization from Lemma 2.12:

$$q_e = \frac{1}{\mathbb{E}N_e(Q_0)}, \quad a_e = A_e(b_e, 1), \quad m_e q_e a_e \asymp 1, \quad q_e \longrightarrow 0. \quad (22)$$

Fix finitely many polygonal quads, one or two replica labels $r \in \{1, \dots, R\}$, and, for each label, finitely many real times. Let F be any function of the corresponding crossing indicators with $|F| \leq 1$. In this section the density at time t is $1/2 + t q_e$. For times in a fixed bounded interval these densities lie in $[0, 1]$ for all sufficiently small e .

5.1 Exact color histories and signed kernels

For a single replica with time vector $\mathbf{t} = (t_1, \dots, t_n)$, a history is an element of $\mathcal{H} = \{0, 1\}^n$. Write $\mathbf{0}$ and $\mathbf{1}$ for its two constant elements and put $P_0 = (\delta_{\mathbf{0}} + \delta_{\mathbf{1}})/2$. If $T \geq \max_i |t_i|$, define the signed measure

$$w_{\mathbf{t}} = \int_{-T}^T \delta_{(\mathbf{1}_{\{s \leq t_i\}})_{i=1}^n} ds - T\delta_{\mathbf{0}} - T\delta_{\mathbf{1}}. \quad (23)$$

This measure is independent of the chosen T once T is large enough. It has total mass zero and total variation at most $4T$. The law of $(\mathbf{1}_{\{U \leq 1/2 + t_i q_e\}})_{i=1}^n$, for a uniform mark U , is exactly

$$P_0 + q_e w_{\mathbf{t}}. \quad (24)$$

For example, in increasing time order the masses of $w_{\mathbf{t}}$ at the all-one and all-zero histories are respectively $\min_i t_i$ and $-\max_i t_i$; its other nonzero masses are successive time differences. Formula (23) also handles tied times without a choice of ordering. It shows that

$$\|w_{\mathbf{t}} - w_{\mathbf{t}'}\|_{\text{TV}} \leq 2 \sum_i |t_i - t'_i|. \quad (25)$$

Thus the finite labeled history space can be kept fixed while the times vary. Different replicas may have different time lists.

Choose a fixed box B containing every tested quad with a positive margin. In the Poisson model choose a geometry event Γ_e which ensures that all cells meeting the tested region have their nuclei in B and that their relevant portions are determined by the nuclei in B . A finite covering by occupancy squares inside B gives such an event, increasing under addition of nuclei, with

$$\mathbb{P}(\Gamma_e^c) \leq C \exp(-c/e^2). \quad (26)$$

We multiply the test by $\mathbf{1}_{\Gamma_e}$; we do not condition on Γ_e . Off this event its value is defined to be zero, so it is a function of the colors of the finitely many sites in B . In the triangular model the fixed margin makes this localization deterministic for all sufficiently small meshes, and we take Γ_e to be the whole sample space.

The counting volume, which converges to planar area, is

$$\sigma_e(dx) = \begin{cases} dx, & \text{in the Poisson model,} \\ m_e^{-1} \sum_{x \in T_e} \delta_x(dx), & \text{in the triangular model.} \end{cases} \quad (27)$$

For an ordered list $\mathbf{x} = (x_1, \dots, x_k)$ of distinct sites in B , choose nonempty sets $I_i \subseteq \{1, \dots, R\}$. For every $r \in I_i$ let $w_{i,r}$ be a signed measure of total mass zero on that replica's history space. Define

$$K_e(F; \mathbf{x}, \mathbf{I}, \mathbf{w}) = \mathbb{E} \left[\int \mathbf{1}_{\Gamma_e} F(\omega[\mathbf{h}]) \prod_{i=1}^k \prod_{r \in I_i} dw_{i,r}(h_{i,r}) \right]. \quad (28)$$

Here, in the Poisson case, $x_1, \dots, x_k$ are first added as nuclei to an ordinary Poisson process; in the triangular case the tiling is unchanged. All unselected site–replica coordinates have independent fair, constant histories. At the selected coordinates $\omega[\mathbf{h}]$ uses the histories being integrated. The expectation averages the remaining fair bits and, in the Poisson case, the Poisson process. An unselected replica at an inserted nucleus is therefore still fair. Insertion of a nucleus in this formula is the Mecke representation of a sum over existing sites; the signed operation itself changes colors and does not define geometric pivotality.

Write $B_{\neq}^k$ for the set of distinct tuples, and $d(\mathbf{I}) = \sum_i |I_i|$. Multiplying (24) over sites and replica labels, and then using the Poisson factorial moment formula, gives

$$\mathbb{E}[\mathbf{1}_{\Gamma_e} F] = \mathbb{E}_0[\mathbf{1}_{\Gamma_e} F] + \sum_{k \geq 1} \frac{m_e^k}{k!} \sum_{\substack{\emptyset \neq I_i \subseteq \{1, \dots, R\} \\ 1 \leq i \leq k}} q_e^{d(\mathbf{I})} \int_{B_{\neq}^k} K_e(F; \mathbf{x}, \mathbf{I}, \mathbf{w}) d\sigma_e^{\otimes k}(\mathbf{x}). \quad (29)$$

In this identity $w_{i,r} = w_{\mathbf{t}(r)}$, and $\mathbb{E}_0$ uses fair constant histories everywhere. The left side uses the actual near-critical histories. For each fixed e the lattice expansion is finite. The Poisson identity and its initial interchange of expectation and summation follow from the exponential moments of the number of nuclei in B . The stronger, mesh-uniform summability needed for taking limits is the content of the next proposition.

Proposition 5.1 (Uniform bound for the site expansion). *Fix a box B , finitely many polygonal crossing tests, and $R \leq 2$ replica labels. Let $g \in (1, 2)$ be an exponent in the lower four-arm bound of Lemma 2.1. There is a constant C , depending only on this data, such that, for every $k \geq 1$, every choice of nonempty I_i , and centered weights $w_{i,r}$,*

$$\frac{(m_e q_e)^k}{k!} \int_{B_{\neq}^k} |K_e(F; \mathbf{x}, \mathbf{I}, \mathbf{w})| d\sigma_e^{\otimes k}(\mathbf{x}) \leq C^k k^{-(1-g/2)k} \prod_{i,r \in I_i} \|w_{i,r}\|_{\text{TV}}, \quad (30)$$

uniformly for all sufficiently small meshes and all $|F| \leq 1$ of the specified crossing indicators. Consequently the absolute sum in (29) is bounded by $\sum_{k \geq 1} C_T^k k^{-(1-g/2)k} < \infty$ when all times have absolute value at most T . For two replicas, the total absolute contribution of terms with $|I_i| = 2$ for at least one i is $O_T(q_e)$.

Proof. We first establish a pointwise geometric bound and then integrate it. Put $W(\mathbf{w}) = \prod_{i,r \in I_i} \|w_{i,r}\|_{\text{TV}}$; the assertion is immediate if one of these factors is zero. For fixed geometry and unselected fair bits, the signed integral in (28) vanishes if F is independent of any one selected history, because its weight has mass zero. Thus a necessary condition for a nonzero integral is the following: at every selected coordinate there are settings of the inserted histories for which changing that coordinate changes at least one tested crossing. The settings witnessing this condition may differ from coordinate to coordinate. We use only the resulting arm tests in regions containing no inserted site, where all the colors are the original fair bits. In particular, no enumeration of all global history assignments is required.

Our geometric objective is one four-arm factor a_e for each inserted site, with the remaining cost expressed as a product of integrable singularities in intersite distances and distances to polygon

vertices and marks. The merging forest will select the distances and the disjoint annuli that supply these factors.

The merging forest. Organizing nearby sites into merging clusters and assigning arm events to the annuli between successive scales has a methodological predecessor in the spectral-sample analysis of Garban, Pete, and Schramm [11, Sections 4.2–4.3]. Here we need estimates for signed history kernels rather than squared Fourier coefficients, and give the required argument. Place fixed roots at all polygon vertices and marks, identifying coincident roots, and denote their number by J . Identify these roots with a common sink for the purpose of Kruskal’s algorithm. On the remaining graph use the ordinary Euclidean distances between all insertion sites and roots, clipped below at $b = b_e$. Starting at scale b , accept an edge when its endpoints are in distinct components, and continue through a fixed scale $L < 1$. Ties are resolved by a fixed order. The constant L is chosen small enough, in terms of the fixed polygons, that balls of radius $O(L)$ do not meet two nonincident sides of any one polygon. The constants in the probes below will also be chosen in terms of the incident angles. Record the ordinary accepted edges, with clipped lengths ℓ_f ; the virtual identification of the roots is not a recorded edge.

A component containing the sink is called grounded. A *life* is a maximal interval $[u, v]$ during which an ungrounded component has the same set of insertion sites. Each merge ends the participating ungrounded lives and, if the merged component is ungrounded, starts a new life. A remaining life ends at L . Zero-length lives caused by ties can be discarded. There are at most $2k$ lives and at most k recorded edges: every accepted edge decreases the number of ungrounded components by one. The ordinary recorded graph is a forest and each of its components contains at most one root, since an edge joining two rooted components would already make a cycle through the sink.

For a life C on $[u, v]$, retain one representative site x_C and write D_C for the diameter of its site set. Take a buffered annular probe centered at x_C , with radii

$$C_0 \max(u, D_C) \quad \text{and} \quad c_0 v, \tag{31}$$

where C_0 is sufficiently large and c_0 sufficiently small. If these radii are reversed, impose no probe and assign it cost one. We use the convention $A_e(r, R) = 1$ when $r \geq R$.

The necessary influence of the representative site gives an arm test on every nonempty probe. The changing cell is localized inside its inner boundary, and the other inserted histories do not enter the probe, on the occupancy events of Lemma 2.3. For each tested quad separately the probe meets no side or at most one straight side: all vertices and marks are outside the outer buffer, and the constants are chosen in terms of the finitely many polygon angles. When a side is encountered partway through the probe, use four arms up to the distance to that side and three half-plane arms thereafter. Lemma 2.2 and the straight-side estimate bound this cost by

$$C_1 A_e(C_0 \max(u, D_C), c_0 v). \tag{32}$$

A purported influence outside the quad is either impossible after localization or has the same straight-side test. Different polygons may overlap; the argument just chooses one quad and one replica for the influence at this probe. There are only a fixed number of such choices per life. Even if its witness uses different times, the bits on the probe are critical and constant in time.

The probes can be chosen with disjoint buffers. If a life D descends from a life C , then x_D belongs to the site set of C and $v_D \leq u_C$. Its outer buffer is thus within distance $D_C + O(c_0 u_C)$ of x_C , strictly inside the inner boundary of C ’s probe when C_0 is large. For two components with

incomparable lives, let v be the larger of their death scales. Their site sets are at distance at least v : otherwise Kruskal would process an edge between them before scale v , forcing the component with that death scale either to contain a site of the other component already at its birth or to change its site set before its death. If the other component has become grounded, this edge would instead ground it before its death. Each alternative contradicts the definition of the life. Their outer radii are at most $c_0 v$, so their buffers are disjoint after decreasing c_0 . This also applies when the earlier component has become grounded. The argument concerns the entire component sets, so a change of representative causes no loss.

In Poisson percolation each localized probe uses only the Poisson points and bits in its buffer. Forced nuclei all lie outside these buffers. If an occupancy test fails, use the outermost-failure decomposition of Lemma 2.3: omit the failed inner part and retain the arm probe beyond it, with disjoint margins. Its alternatives remain measurable in the original probe buffer, and their sum has the same upper bound (32). This also accounts for a cell reaching far from its nucleus. Thus the probe costs multiply. On the lattice their site sets are disjoint when the constant in b_e is sufficiently large. For common-geometry replicas only one replica is used in each probe; disjoint spatial buffers still give the required independence. We obtain

$$|K_e(F; \mathbf{x}, \mathbf{I}, \mathbf{w})| \leq C_2^k W(\mathbf{w}) \prod_{C \text{ a life}} A_e(C_0 \max(u_C, D_C), c_0 v_C). \quad (33)$$

Diameter losses and telescoping. The component diameters in (33) must be removed without a factor growing faster than exponentially in k . Quasi-multiplicativity and the power lower bound show that replacing the factor for a life by $A_e(u_C, v_C)$ costs at most a fixed constant times

$$\exp\left(g \int_{u_C}^{v_C} \mathbf{1}_{\{t < D_C\}} \frac{dt}{t}\right).$$

The fixed factors C_0, c_0 cost another constant per life, including the case of an empty probe. Each ungrounded component is joined by its internal recorded edges, so $D_C \leq \sum_{f \text{ internal to } C} \ell_f$. At a time $u_C < t < v_C$ those edges all have $\ell_f \leq t$. The currently live ungrounded components are disjoint; hence an edge is charged by at most one life at any time. Consequently

$$\sum_C \int_{u_C}^{v_C} \mathbf{1}_{\{t < D_C\}} \frac{dt}{t} \leq \sum_C \sum_{f \text{ internal to } C} \int_{u_C}^{v_C} \frac{\ell_f}{t^2} dt \leq \sum_f \int_{\ell_f}^{\infty} \frac{\ell_f}{t^2} dt \leq k. \quad (34)$$

Clipping edges below b preserves the diameter inequality, so this estimate includes clusters of arbitrarily close insertions.

Set $p_e(t) = A_e(t, L)$. Up to a constant for each life, $A_e(u, v) = p_e(u)/p_e(v)$. Initially there are k factors $p_e(b)$. When two ungrounded components merge at scale ℓ , two denominators $p_e(\ell)$ are canceled by one numerator, leaving one inverse factor. When an ungrounded component joins the sink there is again one inverse factor. Lives surviving to L end with $p_e(L) = 1$. There are therefore no extra terminal factors, and

$$\prod_C A_e(u_C, v_C) \leq C_3^k A_e(b, L)^k \prod_f A_e(\ell_f, L)^{-1} \leq C_4^k a_e^k \prod_f \ell_f^{-g}. \quad (35)$$

The comparison of $A_e(b, L)$ with a_e uses the fixed positive scale L . Combining (33)–(35) proves the required pointwise bound. No estimate is needed for the color law inside a cluster of nearby insertions.

Integrating the forest. Put $s = k^{-1/2}$. Call a recorded edge short when $\ell_f < s$. There are at most k recorded edges. If there are j short edges, the remaining factors in (35) are at most $k^{g(k-j)/2}$; this remains an upper bound when the forest has fewer than k edges. The short edges form a forest with at most one root in each component. We may overcount their choices by $\binom{k+J}{j}^2$, ignoring all conditions imposed by Kruskal's algorithm.

Uniformly in a point y of the plane, the one-edge integral satisfies

$$\int_B (|x - y| \vee b)^{-g} \mathbf{1}_{\{|x-y| \vee b < s\}} d\sigma_e(x) \leq C_5 s^{2-g}. \quad (36)$$

For area measure this follows by polar integration. For lattice volume, $\sigma_e(B(y, r)) \leq C(r^2 + e^2)$, and $b \geq ce$ permits a dyadic-shell summation from b to s with the same bound. This is uniform also when y is a fixed root off the lattice. If $b \geq s$ the integral is zero. Allowing equal lattice sites only enlarges this bound; the atom at $x = y$, when present, has mass $O(e^2)b^{-g} = O(b^{2-g})$.

Integrate the leaves of every short-edge tree. Each edge gives the factor in (36); a tree without a fixed root leaves one free center, whose integral is at most a constant depending on B . A rooted tree leaves no such integration. We conclude that

$$\begin{aligned} \frac{1}{k!} \int_{B^k} \prod_f \ell_f^{-g} d\sigma_e^{\otimes k} &\leq \frac{C_6^k}{k!} \sum_{j=0}^k k^{g(k-j)/2} \binom{(k+J)^2}{j} s^{(2-g)j} \\ &= \frac{C_6^k k^{gk/2}}{k!} \sum_{j=0}^k \binom{(k+J)^2}{j} k^{-j} \leq C_7^k k^{-(1-g/2)k}. \end{aligned} \quad (37)$$

For the last inequality, extend the sum to the binomial expansion of $(1 + 1/k)^{(k+J)^2} \leq C_J^k$ and use $k! \geq (k/e)^k$. Thus both the number of possible forests and the small-scale singularities have been included in the factorial bound.

Finally $(m_e q_e) a_e$ is bounded by (22), giving (30). For $R \leq 2$ there are at most 3^k label assignments. On a bounded time range all variation norms are bounded, and $q_e^{d(\mathbf{I})-k} \leq 1$ once $q_e \leq 1$. The asserted summable majorant follows. If some $|I_i| = 2$, this last factor is at most q_e . The same summable majorant, multiplied by q_e , bounds the entire repeated-label contribution. $\square$

5.2 Location cutoffs and fixed coefficients

The coefficient estimate sums over all locations, including clusters and positions near polygon edges. For the local comparison in the next section we need to remove such positions at arbitrarily small cost. A slightly stronger integrability estimate provides this.

Lemma 5.2 (Uniform integrability in the insertion locations). *Fix k , the polygonal test family, $R \leq 2$, and an upper bound on all the variation norms in (28). There are $p > 1$ and $C_k < \infty$ such that, after extending the kernel by zero to coincident tuples,*

$$\int_{B^k} \left| (m_e q_e)^k K_e(F; \mathbf{x}, \mathbf{I}, \mathbf{w}) \right|^p d\sigma_e^{\otimes k}(\mathbf{x}) \leq C_k. \quad (38)$$

Let Z be the union of all tested polygon boundaries and ∂B . For $\rho > 0$ let $B_\rho \subset B^k$ consist of tuples for which $|x_i - x_j| < \rho$ for some $i \neq j$ or $\text{dist}(x_i, Z) < \rho$ for some i . Then

$$\lim_{\rho \downarrow 0} \limsup_{e \downarrow 0} (m_e q_e)^k \int_{B_\rho} |K_e(F; \mathbf{x}, \mathbf{I}, \mathbf{w})| d\sigma_e^{\otimes k}(\mathbf{x}) = 0. \quad (39)$$

The conclusions are uniform in the indicated tests and weights. They remain true after omitting the geometry cutoff Γ_e from the definition of the kernel.

Proof. Choose $p > 1$ with $gp < 2$. For fixed k , there are only finitely many ordinary forests on the k sites and the J fixed roots. The pointwise estimate (35), multiplied by $(m_e q_e)^k$, is bounded by a constant times the sum over all such forests having at most one root per component of $\prod_f (|x_{i(f)} - x_{j(f)}| \vee b)^{-g}$, with a root coordinate interpreted as fixed. Raising this finite sum to the power p costs only a constant depending on k . Leaf integration, now with exponent $gp < 2$ and with distances bounded by a fixed constant containing B and all roots, bounds each forest integral uniformly in e . The lattice estimate is the same dyadic calculation as (36). This proves (38).

Hölder's inequality bounds the expression before the limits in (39) by $C_k^{1/p} \sigma_e^{\otimes k}(\mathcal{B}_\rho)^{1-1/p}$. For area measure this volume tends to zero as $\rho \downarrow 0$. For lattice volume its mesh limit is the same: the finitely many polygon edges and the pairwise diagonals have zero area in the relevant product space, and their ρ -neighborhoods can be bounded by finite unions of boxes with arbitrarily small total volume as ρ decreases. This proves the ordered-limit assertion and includes vertices, marks, and box edges.

It remains to justify removal of Γ_e . Adding a fixed number of nuclei can only improve its occupancy requirements. Thus the difference between truncated and untruncated kernels is bounded uniformly in the locations by $C_k \exp(-c/e^2)$. The normalization $(m_e q_e)^k$ grows only polynomially in $1/e$, by the lower arm bound in Lemma 2.1. The normalized difference therefore converges uniformly to zero, also in L^p . $\square$

For later calibration, consider a single time and the centered weight $\Delta = \delta_1 - \delta_0$, enlarging B to contain Q_0 with a margin if necessary. For the increasing crossing test of Q_0 the corresponding one-site kernel is its nonnegative color-pivotal probability. Russo's identity and the definition of q_e give

$$(m_e q_e) \int_B K_e(\mathbf{1}_{C_{Q_0}}; x, \Delta) d\sigma_e(x) = 1 + o(1). \quad (40)$$

Here the single replica label is suppressed. Without the box cutoff, the integral over all sites is exactly $q_e \mathbb{E} N_e(Q_0) = 1$. For the displayed truncated version its difference is negligible: localization gives exponentially small failure probability and the number of cells meeting a fixed compact set has moments growing at most polynomially in $1/e$, as in Lemma 2.12. Cauchy-Schwarz then also controls the lost pivotal count.

We have reduced all finite-time tests to an absolutely convergent series of critical signed kernels. Terms using both replicas at a single site disappear, and each remaining fixed-degree integral may be restricted to compact sets of separated locations away from the polygon boundaries. These are precisely the conditions under which the local comparison in the next section will identify its limit.

6 Replacing a site by a mesoscopic square

The expansion in the preceding section reduces the problem to identifying signed kernels at finitely many separated insertion points. A microscopic insertion is not itself a continuous critical observable. We compare it with forcing the colors in a square of fixed positive size, which is covered by Proposition 4.5. Approximating microscopic pivotal behavior by mesoscopic crossing data is central to the construction of Garban, Pete, and Schramm [12, Section 4]. Here the comparison must preserve the signs of history weights and allow two color replicas on the same Poisson geometry. This section proves that comparison before taking any limit of microscopic pivotal amplitudes.

6.1 Mixed kernels and the comparison statement

Fix a finite family of polygonal quads, a bounded function F of their crossing indicators at finitely many labeled times, and either one or two color replicas. Assume $|F| \leq 1$. Fix $k \geq 1$ insertion points $\mathbf{x} = (x_1, \dots, x_k)$ in a compact set of configurations on which the points are mutually separated and stay a positive distance from all quad boundaries and from the boundary of the integration box. There is one active replica label at each insertion point, and a signed measure w_i of total mass zero on that replica's finite history space. Its total variation is bounded by a fixed constant. Repeated labels at a single site were disposed of in Proposition 5.1.

A *site insertion*, denoted s , has the meaning used in the expansion: in the Poisson model add a nucleus at x_i and impose the selected history on its cell; on the triangular lattice impose it at the site x_i . A *square insertion*, denoted d , uses the ordinary geometry, without adding a Poisson nucleus, and imposes the selected history on every cell meeting

$$B_\infty(x_i, h) = x_i + [-h, h]^2.$$

Write

$$K_e^t(F; \mathbf{x}, \mathbf{w}), \quad \mathbf{t} \in \{s, d\}^k,$$

for the resulting signed kernel, with the replica labels suppressed. All unspecified colors are fair and constant across the tested times. If two instructions apply to one cell, use the instruction with the smaller insertion index. Localization shows that such conflicts between separated insertion regions have exponentially small probability for fixed h and small e . The same convention applies to forced-square critical tests. We remove the large-box truncation in these fixed-degree kernels; its exponentially small error remains negligible after their normalizations.

Set

$$a_e = A_e(b_e, 1), \quad a_{e,h} = A_e(h, 1), \quad \gamma_e(s) = a_e, \quad \gamma_e(d) = a_{e,h}.$$

Independent localized probes around the insertion points, using Lemma 2.3, give

$$|K_e^t(F; \mathbf{x}, \mathbf{w})| \leq C \prod_{i=1}^k \gamma_e(t_i). \quad (41)$$

Indeed, the centered integral vanishes unless changing each inserted history can affect one of the tested crossings, for some settings of the other histories. Outside its insertion region this implies four alternating arms on the unchanged fair bits. The probes start at a constant multiple of b_e or h and end at a fixed radius smaller than the separation distances. Cells extending out of an insertion region are handled by the outermost failed occupancy test, rather than by assuming that their diameters are bounded. Quasi-multiplicativity compares the probe costs to a_e and $a_{e,h}$. Constants may depend on the fixed compact set and the bounds on the weights, but not on e or sufficiently small h .

Let $G_x = \mathbf{1}_{C_{x+Q_0}}$, where $Q_0 = [-1, 1] \times [-1/2, 1/2]$ has its vertical sides as target arcs. Use one replica and the one-time weight $\Delta = \delta_1 - \delta_0$ at x . Denote the corresponding site and square kernels by

$$S_e = K_e^s(G_x; x, \Delta), \quad D_e(h) = K_e^d(G_x; x, \Delta).$$

Stationarity makes these independent of x ; triangular centers are lattice sites. Monotonicity and (41) give $0 \leq S_e \leq Ca_e$ and $0 \leq D_e(h) \leq Ca_{e,h}$.

Lemma 6.1 (A square can transmit a pivotal change). *There is $c > 0$ such that, for every sufficiently small fixed $h > 0$ and all sufficiently small e depending on h ,*

$$D_e(h) \geq ca_{e,h}. \quad (42)$$

The same assertion holds for the one-time square-insertion kernel of a fixed square or rectangle crossing when its center varies in a sufficiently small open neighborhood of the center of that test. The constant is uniform over that neighborhood.

The proof is given below, after the needed arm-extension construction. Our principal assertion compares products, so that no microscopic amplitude has to be known in advance.

Proposition 6.2 (Signed relative replacement). *Under the preceding assumptions, fix the types $t_2, \dots, t_k$ and put*

$$K_s = K_e^{(s, t_2, \dots, t_k)}(F; \mathbf{x}, \mathbf{w}), \quad K_d = K_e^{(d, t_2, \dots, t_k)}(F; \mathbf{x}, \mathbf{w}), \quad P_e = \prod_{i=2}^k \gamma_e(t_i).$$

The empty product is 1. In either model,

$$\lim_{h \downarrow 0} \limsup_{e \downarrow 0} \sup_{\mathbf{x}} \frac{|K_s D_e(h) - K_d S_e|}{P_e a_e a_{e,h}} = 0. \quad (43)$$

The supremum is over the fixed compact separated range, restricted to lattice sites in the triangular model; in Poisson insertion integrals, null general-position exceptions are omitted. The assertion is uniform for bounded total variations of the centered weights.

We prove the proposition by joining two independent colored geometries through one of finitely many annuli around x_1 . The joining annulus reduces the information transmitted from its inside to a single bit. Several trials are necessary: an absolute error tending to zero would not suffice in (43), whose denominator also tends to zero.

6.2 Guarded arms and four-run cycles

All annuli in the next two lemmas have positive radii fixed before the mesh limit. A local geometry condition is called typical if its probability tends to one as $e \downarrow 0$. Such conditions will always be measurable in a specified enlarged annulus. They include occupancy screens, a mesh bound there, and lower bounds for the finite lists of rectangle-chain crossing probabilities used in that annulus. Proposition 2.4 supplies the crossing conditions. The arm concentration input in Lemma 2.1 also supplies quenched arm bounds at these fixed radii. With spare constant-factor margins, these are bounds for proper disjoint site paths between graph cycles approximating the prescribed curves. For lower bounds, start with an arm request on a slightly longer annulus; for upper bounds, transfer a graph request to a slightly shorter one. Localization and quasi-multiplicativity absorb these changes of radii.

A *guarded endpoint* of a path consists of a small box reached by the path, a monochromatic circuit surrounding that box in a larger concentric box and connected to the path, and a reserved further enlargement that every other retained arm is required to avoid, including an arm of the same color. A connector beginning inside the smaller box must meet the surrounding circuit. All boxes have sizes that are fixed positive fractions of the relevant annular end scale. Their positions come from a finite deterministic list. The following construction uses the classical arm-separation and RSW gluing method [15, 16]; we specify its coordinate supports because a subsequent signed comparison needs more than an unconditional lower bound.

Lemma 6.3 (Alternating arms with guarded landings). *Fix sufficiently large end margins and four separated target patches at each end of an annulus of scales $l < L$, in matching alternating cyclic order. Assume L/l exceeds a constant determined by these end layouts and margins. The ratio may be arbitrarily large but is fixed before the mesh limit. On typical local geometries, the conditional probability of four alternating arms with guarded endpoints in those patches is at least*

$$c A_e(l, L). \quad (44)$$

The constant depends on the prescribed end layouts and margins, but is independent of the intermediate ratio. In the endpoint overlap regions, requirements use each color coordinate monotonically in its prescribed color order, with no opposite-color requirement on the same coordinate. The entire event is measurable in the annulus with its end margins.

Proof. Choose deterministic graph trim cycles a fixed number of doublings inside the two end collars, leaving a factor of room on both sides. Use the quenched alternating-arm lower bound on a longer annulus. A bad trim means that two distinct proper arms, or an additional visit of another arm, approach within relative distance δ of a candidate endpoint. Both relevant path pieces continue to the two sides of the collar. Lemma 2.9 bounds the probability of such an event in the collar by a quantity tending to zero with δ : cover the trim curve by $O(\delta^{-1})$ balls, and use the any-color four-arm bound of Lemma 2.7 in each ball.

This absolute collar estimate yields a relative error here. Retain the necessary alternating arms in the rest of the original annulus, deleting only a fixed number of end scales and a buffer at the bad collar. Its color coordinates are disjoint from those of the collar test. Multiplying the two bounds and using quasi-multiplicativity gives at most $o_\delta(1)A_e(l, L)$, uniformly in the intervening ratio. Choose δ so that the two bad ends together cost less than half the arm lower bound. The retained proper arms then end in four boxes from a fixed finite list; only the appropriate arm enters the reserved enlargement of each box. Loop erasure and first-contact trimming keep retained paths on their designated side of the trim, up to the vanishing mesh margin.

For each choice of endpoint boxes, apply Lemma 2.10. A lasso surrounds each endpoint box, a corridor starts inside that lasso, and disjoint corridors route in cyclic order to the target patches. A cyclic offset is allowed. These routes use only the fixed-ratio end spaces, so their widths and crossing costs do not depend on L/l . Their overlaps with retained paths occur solely in the reserved color-specific enlargements. Conditioning on coordinates outside the connector supports, Harris' inequality in those supports gives a positive factor times the retained arm event. The event being extended is the existence of suitable paths, not a conditioning on a previously inspected choice of paths. Taking a maximum over the finite endpoint choices proves (44) and the asserted support and monotonicity properties. $\square$

Proof of Lemma 6.1. Apply Lemma 6.3 between scales comparable to h and a fixed macroscopic radius about the square center. At the small end continue the four separated tracks into four patches lying strictly inside $B_\infty(x, h)$. At the large end continue the black tracks across the target sides of the rectangle and the white tracks across the other two sides, away from its corners. These end extensions have uniformly positive conditional probability on typical geometry. Forcing the square black joins its two black tracks and gives the tested crossing. Forcing it white joins its white tracks and gives a complementary white crossing with clearance from black. Thus the square changes the crossing. The event has conditional probability at least $cA_e(h, 1)$ on typical geometry. Averaging, and taking the mesh sufficiently small for the fixed h , proves (42). All tracks can be chosen with fixed positive widths for centers in a small central neighborhood, giving the final uniform assertion. $\square$

A *four-run cycle* is a simple site cycle surrounding the inner boundary of an annulus whose cyclic color word has exactly four nonempty runs, alternately black and white. We also call it a necklace. The run vertices are included when testing connections on its inside.

Lemma 6.4 (A sleeve containing a necklace). *There is a fixed sufficiently large annulus ratio for which the following event has conditional probability at least $c_0 > 0$ on typical local geometry: four alternating arms cross the annulus, have guarded endpoints in prescribed patches at both ends, and meet a necklace in its middle, one arm through each corresponding run. All additional requirements beyond the guarded arm event are supported away from the endpoint overlap regions.*

Proof. Leave a core annulus strictly between the end regions, of ratio L/l which will be chosen large. Guarded alternating arms through the sleeve cost at least $c(l/L)^g$, where $g < 2$, by Lemma 6.3 and the power lower bound. Exclude in the core four alternating arms together with a fifth disjoint arm of either color. The quenched upper bound for this event on typical geometry is at most $C(l/L)^2$. The end margins change these bounds only by fixed constants. Choosing L/l large and then fixing it leaves positive probability, uniformly for small meshes. The exclusion uses only core coordinates and therefore preserves the endpoint monotonicities.

We verify that the exclusion produces the claimed cycle. Trim the four arms to proper simple traversals of the core. They divide it into four topological rectangles, each with an opposite-colored arm as each of its two lateral boundaries. In such a gap, let $\mathcal{B}$ be the black cluster reached from its black lateral boundary, using only vertices of the gap. If $\mathcal{B}$ is adjacent to the white cluster of the other lateral boundary, there is a transverse path that is first black and then white, with a single color switch. Otherwise let $N(\mathcal{B})$ be the external vertex boundary of $\mathcal{B}$ in the gap. Every vertex of $N(\mathcal{B})$ is white, and every transverse path meets it. Temporarily color $N(\mathcal{B})$ white and its complement black. Hex duality in the gap then makes $N(\mathcal{B})$ contain a proper white path between the two annular ends, disjoint from the white lateral boundary and from the four original arms. This is the excluded fifth arm. Hence the single-switch transverse path exists in every gap. Orient it from its black to its white side, stop at its first white-side contact, and start at the last preceding black-side contact. Erase loops within each color segment. Its interior then avoids both original side arms.

Join the two contacts on each original arm by the intervening subpath of that arm. The four transverse paths lie in disjoint gaps, and these four joining subpaths lie on disjoint arms. Their union is a simple enclosing cycle; if two contacts on an arm agree, the joining subpath is a single vertex. Its only color changes are the four switches on the transverse paths. It therefore has the required four runs. Each original arm meets its corresponding run, and its portions on either side supply the connections to the two guarded ends. $\square$

6.3 A successful seam transmits a single bit

To represent $K_s D_e(h)$, take independent environments ω^A and ω^B . An environment contains a full-plane Poisson process and fair bits of both replicas at every nucleus, or just the fair bits at all sites in the triangular model. Give unused replica labels fair bits as well. The first environment carries the F test and its insertions; the second carries G_{x_1} . Use in G_{x_1} the same replica label as the first insertion of F , called the *active label*. Added nuclei and the insertion types are deterministic annotations of the environment, separate from its Poisson process. An annotated nucleus is supplied with independent fair bits for every replica, including a label unused by its insertion; these bits travel with the annotation. Histories at inserted regions are summed with the weights of the tests, rather than included in these fair-bit environments.

Fix a deterministic circular seam centered at x_1 , outside both possible first insertion regions, whose closed disk is disjoint from all other insertion regions and all tested quad boundaries. Its trial annulus, including the inner and outer sleeves described below and all buffers, is chosen in a disk with the same disjointness properties. Call the closed disk extending to this trial's outer boundary the *trial disk*. Exchanging the insides means exchanging the Poisson points and both fair bits at points inside that circle, together with the annotation specifying the first insertion. On the lattice only the bits and insertion type are exchanged; choose a deterministic side for a site on the seam. There are four combinations of an inside from A or B and an outside from A or B .

Choose an inner sleeve lying strictly inside the seam and an outer sleeve lying strictly outside it. The seam must transmit connections both before and after exchanging the insides. We therefore require the same construction to work for all four combinations of a source inside and a source outside. The *successful seam event* has the following requirements. Occupancy and mesh screens localize the two sleeves and the joining region, in all four combinations. They make the inner sleeve depend only on its designated inside and the outer sleeve only on its designated outside, and screen them from the center and from all other insertions. Each of the two inner sleeves and two outer sleeves has an active-label necklace. There is a choice of these four necklaces such that every inside–outside combination has four disjoint alternating connections between its two necklaces, meeting distinct corresponding runs in cyclic order. All these requirements are measurable in one enlarged trial annulus. In particular they use no history setting at any insertion point. They are symmetric in the two sources: exchanging the insides permutes the four combinations and preserves the event. One may keep all admissible necklace tuples. When comparing a configuration with its exchange, use the same source-indexed tuple, transporting its inner necklaces with the exchanged insides and leaving its outer necklaces fixed. The transported tuple is admissible after the exchange, so the success event needs only the existence of a tuple.

Lemma 6.5 (The information at a necklace). *On a successful seam event, let $b \in \{0, 1\}$ record whether the two black runs of its inner necklace are connected inside that necklace. The same connection test at its outer necklace equals b . For every active-label crossing whose quad contains the entire closed trial disk, the dependence on the interior is only through b . A quad disjoint from the trial disk is unaffected. These are the only two cases for the fixed tests in the construction.*

Proof. Hex duality in the disk bounded by the inner necklace says that precisely one of the two black runs being joined or the two white runs being joined holds. Connections include the vertices of the runs. The four alternating arms transmit whichever join holds to the corresponding runs of the outer necklace. The opposite join at the outer necklace is impossible by planar separation. Thus the bit is preserved.

For the last assertion cut the graph at the outer cycle. Each run is already connected along the cycle, so the only undetermined same-color connection among boundary vertices is the join of the two black runs, or equivalently the alternative white join. An exterior path can therefore be read from this bit and the fixed exterior graph. The cycle is separated from every tested boundary. Site paths and cell paths agree at this cut: an inner site cannot connect to the exterior without meeting a cycle site, and the cycle cells and their stars are localized by the screens. Outside that cut the original closed-cell convention is left unchanged. A quad disjoint from the inner disk cannot use it. The same argument permits arbitrary fixed histories at the other insertion regions, which lie outside the trial. $\square$

For a fixed fair-bit environment, the connection bit at an inner necklace is a monotone Boolean function ϕ of the bit imposed at the first insertion. This holds for both a single site and an entire

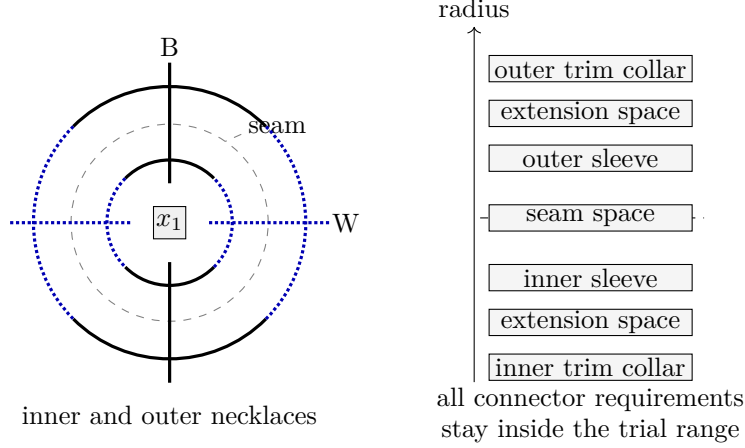


Figure 3: The sewing construction, drawn schematically with round cycles. Solid black and dotted blue denote the two colors. Four ordered arms carry the connection bit between necklaces across the deterministic seam. The right-hand diagram gives the radial order of one trial; each displayed space has fixed multiplicative width, with additional buffers between supports. Actual necklaces are site cycles, and the central forced region is a square.

square. Thus ϕ is constant or the identity. At several times the same function acts coordinatewise on the inserted history, because all other bits inside the necklace are constant across time.

Fix the two environments and all histories at the other insertions. For now assume that every inactive-label crossing, if present, is unchanged by the exchange; we establish this protection below. Let $f_A(\mathbf{h})$ be the value of the exterior F test when its outer necklace receives history $\mathbf{h}$, and let $g_B(b)$ be the value of the exterior G test when its outer necklace receives bit b . The necklace lemma makes these exterior functions independent of which inside supplies the transmitted bit. Write $a_A = \phi_A(1) - \phi_A(0)$ and $a_B = \phi_B(1) - \phi_B(0)$, so $a_A, a_B \in \{0, 1\}$. Since both weights are centered, the product of the signed responses before the exchange is

$$\begin{aligned} & \left[\int f_A(\phi_A \circ \mathbf{h}) dw_1(\mathbf{h}) \right] \left[\int g_B(\phi_B(b)) d\Delta(b) \right] \\ &= a_A a_B \left[\int f_A(\mathbf{h}) dw_1(\mathbf{h}) \right] \left[\int g_B(b) d\Delta(b) \right]. \end{aligned} \tag{45}$$

Here $\phi_A \circ \mathbf{h}$ means coordinatewise composition. If a response function is constant, centering annihilates its factor; otherwise it is the identity. Exchanging the insides exchanges ϕ_A and ϕ_B , while f_A, g_B and their weights stay with the exterior tests. The right-hand side is unchanged. This is a pointwise equality after the signed history sums, not an equality between arbitrarily paired raw histories. Summing the other histories therefore preserves it.

6.4 The adaptive exchange and relative error estimates

Choose finitely many trial annuli with disjoint enlarged supports and a seam in each. All lie beyond both first-insertion scales. Let A_j be the event that trial j is the first successful trial in a fixed order, and let A_0 be the event that none succeeds. If $\mathcal{T}_j$ exchanges the insides at trial j , define

$$\mathcal{T}(\omega^A, \omega^B) = \begin{cases} \mathcal{T}_j(\omega^A, \omega^B), & \text{on } A_j, j \geq 1, \\ (\omega^A, \omega^B), & \text{on } A_0. \end{cases}$$

Each fixed exchange $\mathcal{T}_j$ preserves the product law: restrictions of two independent marked Poisson processes to the inside are independent and identically distributed, as are their fair bits. On the lattice the same observation applies to the bits. The added nucleus is an annotation that moves from the site environment to the other environment; no Poisson intensity or Palm weight is changed.

The success status of trial j is invariant under $\mathcal{T}_j$. A trial farther inside sees its two source environments exchanged wholesale, and a trial farther outside is unchanged. The success test is symmetric in its sources, so every trial's status, and therefore A_j , is invariant under $\mathcal{T}_j$. The sets $A_0, A_1, \dots$ partition the environment space; applying a measure-preserving involution on each invariant piece proves that $\mathcal{T}$ is itself a measurable measure-preserving involution on the underlying unannotated marked environments. On its successful set it also exchanges the site-square and square-site annotations. On A_0 the annotations stay in place, and we estimate that part separately.

We now estimate the error relative to the necessary four-arm probabilities. First we remove failures of the microscopic screens. An outer annulus then protects the inactive replica, while many smaller annuli provide trials for the active label. The square radius h will be taken below all these trials only after their number and positions have been fixed. Choose a fixed macroscopic probe radius R sufficiently small for the separations and tested boundaries. Necessary probes at x_i , $i \geq 2$, have disjoint buffered supports from all the constructions at x_1 ; they always contribute at most CP_e . We therefore estimate only the two environments at x_1 .

Fix a large constant M . By the outermost-failure estimate of Lemma 2.3, configurations contributing to either signed product with an occupancy failure at a scale at least Mb_e in its site probe have total probability at most

$$\varepsilon(M)a_e a_{e,h}, \quad \varepsilon(M) \longrightarrow 0. \quad (46)$$

For the square probe, failures above a constant multiple of h contribute $o_e(1)a_e a_{e,h}$ at each fixed h . These estimates do not assume arms on failed shells: group by the outermost failure, retain the necessary probe beyond it with a disjoint buffer, and apply the power lower bound and quasi-multiplicativity to sum the lost inner scales.

Outside these errors a necessary event E consists of the occupancy screens and four alternating active-label graph arms in both environments, from scales comparable to Mb_e and h , respectively, to a scale comparable to R . Use deterministic graph-cycle cuts with room at the ends. These arms use the base fair bits: no inserted history reaches any of their screened probe regions. Although different crossing coordinates may witness influence, the first insertion can influence only its active label, so all these alternatives imply the same event E . In particular

$$\mathbb{P}(E) \leq C(M, R)a_e a_{e,h}. \quad (47)$$

The definition is made separately with the appropriate site and square starting scales on the two sides of the comparison. The event E is used only for upper bounds; the adaptive exchange is defined by its symmetric success tests and does not require E to be invariant.

Two ways of keeping these bounds relative will be used repeatedly. First, let J be a fixed positive-scale intermediate annulus, with buffers, and let H_J be a geometry event measurable in its enlargement whose failure probability tends to zero. Retain the necessary arm probes strictly below and above that enlargement, in both environments. These probes are independent of its geometry. Quasi-multiplicativity bounds their product by $C(M, R)a_e a_{e,h}$ divided by the two fixed-ratio arm costs omitted at J . Those costs have positive mesh-uniform lower bounds. It follows that

$$\mathbb{P}(E \cap H_J^c) = o_e(1)a_e a_{e,h}. \quad (48)$$

This also applies to geometry requirements in a sewn combination: a fixed inside–outside combination has the ordinary Poisson law, and its local geometry is still measurable in the two enlarged source annuli. A finite list of all such requirements is typical. No typicality assertion is being made at the microscopic scale b_e .

Second, choose $l_* > 0$ much smaller than R and put all trial annuli strictly below l_* , with buffers. When two replicas are used, impose the following *protection event* H : screening holds on an annulus from constant multiples of l_* to constant multiples of R , and in neither source environment does the inactive label have four alternating arms through that annulus. It uses only sites above all the seams, hence is invariant under every exchange. If a tested crossing of the inactive label changed after an exchange, the two configurations would agree on this annulus and differ only on its inside. The pivotal arm implication would then give the forbidden four arms there. The argument works for every setting of the other histories, which are all outside the probe, and includes changes of cells caused by the exchanged geometry.

For completeness, conditioning on the common geometry within this probe, the two labels are independent. If $p_e(\eta)$ is its quenched active four-arm probability, the probability of four arms for both labels is $\mathbb{E}[p_e(\eta)^2]$, bounded by $CA_e(l_*, R)^2$ by the second-moment arm estimate. Retain the active probe below it and the necessary probe in the other environment with disjoint buffers. Quasi-multiplicativity then gives

$$\mathbb{P}(E \cap H^c) \leq [C(M, R)A_e(l_*, R) + o_e(1)]a_e a_{e,h}. \quad (49)$$

Sum over which of the two source environments fails protection. Fixed radius changes are absorbed into the constant; mesh failures are covered by (48). The power upper bound makes the bracket as small as desired by choosing l_*/R small and then the mesh small. This extra arm factor is the reason common Poisson geometry for the two replicas is compatible with the exchange argument. With one replica set H equal to the entire probability space.

6.5 Many annular trials

It remains to show that E almost always supplies a successful seam on the relative scale $a_e a_{e,h}$. Each trial has the radial layout in Figure 3: an inner trim collar, an extension space, an inner sleeve, a seam space, an outer sleeve, another extension space, and an outer trim collar. Every part has a fixed sufficiently large ratio. Trim curves run through the middle of their collars; the deterministic seam runs through the middle of its space. The sleeve ratio is that of Lemma 6.4. Include buffers between supports, and choose all endpoint boxes and corridor routes from finite lists scaled with the trial. For any fixed number n , one can place n such similar, buffered trials below l_* at fixed positive scales. Subsequently take h below the innermost trial.

For trial J , let T_J be the typical geometry event giving all its screens, the quenched lower bounds for both sleeve constructions in both environments, and all the rectangle-crossing lower bounds for the corridor lists in every possible sewn combination. Screens in particular make each sleeve independent of the source used on the far side of the seam. Each sewn geometry has the ordinary law, so a finite intersection supplies T_J . Call J *disqualified*, an event denoted B_J , if T_J fails or a bad trim occurs at either of its two trim curves in either environment. The bad trim test is the local close-approach event of Lemma 2.9, including extra visits near the reserved endpoint boxes. Its paths are proper simple arms through a wider collar; thus the two visits giving the test really yield four arms in a buffered neighborhood, regardless of their color order.

For any $\rho > 0$, the trim tolerance and the local geometry requirements can be chosen so that,

for every trial,

$$\limsup_{e \downarrow 0} \frac{\mathbb{P}(E \cap B_J)}{a_e a_{e,h}} \leq \rho. \quad (50)$$

Here and below h is fixed below the trials. For geometry failures this is (48). For a bad trim retain the necessary parts of that environment's probe on both sides of its collar, and the other environment's probe. Their supports, including Poisson buffers, avoid the bad test. The absolute close-approach probability multiplies their arm costs; deleting the fixed-ratio collar costs only a fixed factor by quasi-multiplicativity. Choose its tolerance to absorb that factor. This reasoning is uniform over scaled copies of the trial: ρ and the construction constants do not depend on n or on a trial's radius. The mesh threshold may depend on the fixed finite collection.

Write U_J for the symmetric successful seam event in trial J , and $N = \bigcap_J U_J^c$ for failure of every trial. The trials cannot be treated as independent after restricting to the necessary long-arm event E . Instead, we compare failure of every trial with success of exactly one specified trial. The latter events are disjoint as that trial varies, which will give the gain from the number of trials.

Lemma 6.6 (One-trial comparison). *For the preceding fixed layouts and tolerances, there is $C_* < \infty$, independent of the number of trials, their scales, and sufficiently small mesh, such that*

$$\mathbb{P}(E \cap N \cap B_J^c) \leq C_* \mathbb{P}\left(E \cap U_J \cap \bigcap_{I \neq J} U_I^c\right) \quad (51)$$

for every trial J .

Proof. Condition first on all Poisson geometries, keeping only realizations in T_J and in the geometry part of E . In each source environment retain four arms from the inner beginning of its probe to the inner trim curve, and four arms from its outer end down to the outer trim curve. Stop at first contact from the retained side, with the graph margins prescribed above. Since the trim is not bad, their endpoint boxes can be chosen from a fixed finite list with matching cyclic order and the reserved avoidance conditions. For each choice let $\mathcal{R}$ be the event that such retained paths exist, together with failure of all trials other than J . Delete the color part of disqualification, failure of trial J , and all arm requirements between its two trims. The events $\mathcal{R}$ for the finitely many box and order choices cover the left-hand event of (51). Their definition concerns existence of retained paths; we do not reveal or condition on a selected path.

In the four sleeves, two in each source environment, impose the events of Lemma 6.4, with cardinal endpoint patches of matching alternating colors. Sleeve coordinates are disjoint from the retained path coordinates and from the other trial supports. Thus these four events have joint conditional probability at least c_0^4 , and intersecting them with $\mathcal{R}$ costs at most that fixed factor.

We describe precisely the additional connector supports. In each extension space use disjoint polygonal tracks from the retained endpoint boxes to the corresponding sleeve patches, first leaving the trim in its normal direction and then respecting annular cyclic order. At each end start a track inside the box surrounded by the guarded lasso. In the seam space use four fixed separated tracks, of the same respective colors for every inside–outside combination, and require rectangle-chain connections in each of its four sewn geometries. Every rectangle and lasso used by a connector lies in these tracks or in a reserved endpoint enlargement. Include their cell stars in the supports.

Let V_B and V_W be, respectively, the original source-bit coordinates used by black and white connector requirements, taking the union over the four sewn combinations. The positive widths between opposite-color tracks and the mesh screens ensure $V_B \cap V_W = \emptyset$; a source site cannot belong to two opposite track supports, even if it is used in several sewn geometries. On $V = V_B \cup V_W$ put

the product order that favors black on V_B and white on V_W . Every connector event is increasing in this order. Its conditional probability, with geometries fixed, has a uniform positive lower bound from the crossing conditions in T_J . Harris' inequality therefore bounds the probability of all connector events by a positive constant from below, since their number is fixed. These events depend only on the coordinates in V .

The retained-and-sleeve event has the same monotonicities on V . For retained paths the only overlaps are their reserved, color-specific endpoint enlargements. For sleeves the only overlaps are their guarded end patches; the five-arm exclusion giving the necklace lies strictly in the core and uses no coordinate of V . Requirements that paths avoid other boxes are restrictions on allowed path locations, so favoring their own colors preserves the existence of the required paths. Failures of other trials use no coordinate of V , or of these sleeves, because the enlarged trial supports are disjoint. They need not be monotone.

Now condition on all color coordinates outside V , apply Harris in V to the retained-and-sleeve event and the intersection of the connector events, and integrate back. This costs another fixed positive factor. All joining requirements are realized simultaneously, including those in geometries that combine different sources. Lassos ensure actual connections to retained paths, while the reserved tracks ensure disjointness, including between arms of the same color. In the two original environments the resulting arms complete E . In all four combinations they give U_J , with the common necklaces supplied by the sleeves. Other trials still fail. Hence the probability for each retained choice is at most a constant times the right-hand probability in (51). Summing over the bounded finite choices and then integrating the geometries proves the claim. $\square$

Summing (51) over n trials, its right-hand events are pairwise disjoint. On the left,

$$n\mathbb{P}(E \cap N) \leq \sum_J \mathbb{P}(E \cap N \cap B_J^c) + \sum_J \mathbb{P}(E \cap B_J).$$

Together with (47) and (50), this gives

$$\limsup_{e \downarrow 0} \frac{\mathbb{P}(E \cap N)}{a_e a_{e,h}} \leq \rho + \frac{C_* C(M, R)}{n}. \quad (52)$$

The averaging over disqualifications, rather than a union bound over all of them, is what leaves ρ instead of $n\rho$.

Proof of Proposition 6.2. Choose a desired relative error $z > 0$. First fix M so that the cutoff error in (46) is sufficiently small. Next choose l_*/R so that (49) is sufficiently small. Fix the sleeve and extension ratios, and the trim tolerances so that the disqualification term ρ is sufficiently small. These choices fix C_* . Choose n so large that $C_* C(M, R)/n$ is sufficiently small, and place that many buffered trials below the protection annulus. Only now take h sufficiently small to lie below every trial; for each such fixed h , let $e \downarrow 0$. All local constructions are therefore at positive scales when their geometry conditions are imposed.

On configurations with a successful protected seam, the adaptive measure-preserving involution and the signed identity following Lemma 6.5 identify the two product integrals. The remaining portions are bounded in total variation. A nonzero signed contribution has the necessary probes at every insertion point; the independent probes at the other points contribute CP_e . At the first point the discarded portions are bounded by (46), (49), and (52), together with the vanishing fixed-scale geometry errors. The weights contribute only their fixed total-variation bounds. Thus the limsup of the ratio in (43) is at most Cz ; since z is arbitrary, it is zero.

All constants are uniform on the fixed compact separated location range. Local layouts are translated copies of the same finite lists. In the Poisson model Mecke points at other insertions lie outside their buffered supports, and the screens make those annotations irrelevant there. In the triangular model translations are by lattice sites. This proves the stated uniformity and completes the comparison. $\square$

7 Identification and the quenched limit

The expansion and local comparison now identify the law. At the auxiliary scale q_e , the first derivative of the Q_0 crossing probability is exactly one at criticality. Its square-insertion approximation has a common critical limit, so this identity determines the unknown site-to-square comparison factor. We then identify every finite crossing test, convert from the rectangle to the unit-square normalization, and pass to the conditional law on $\mathcal{K}$. All quantities indexed by e in this section are formed separately in each model, with the notation of Sections 5 and 6.

7.1 Calibration by one crossing derivative

Recall that $q_e = (\mathbb{E}N_e(Q_0))^{-1}$ for the centered auxiliary rectangle Q_0 . For its translate centered at a proposed insertion point, let S_e be the critical single-site kernel and $D_e(h)$ the critical square-insertion kernel, both with weight $\delta_1 - \delta_0$. These are the nonnegative quantities introduced in Section 6. Put

$$c_e = m_e q_e S_e. \quad (53)$$

The arm upper bound on S_e , together with $q_e \asymp (m_e a_e)^{-1}$, shows that c_e is bounded.

Lemma 7.1 (Common calibration). *There is a constant $c \in (0, \infty)$ such that $c_e \rightarrow c$ in both models. Moreover, for each fixed degree, assignment of one replica label to each insertion, and finite signed crossing test, its normalized all-site kernel integral has a limit common to the two models.*

Proof. Fix a compact range of insertion locations separated from one another and from all polygon boundaries. Write $K_{e,s}(x_1, \dots, x_k)$ for the all-site kernel and $K_{e,h}(x_1, \dots, x_k)$ for the kernel with all insertions replaced by squares of radius h . Proposition 6.2, applied successively at the k locations, gives

$$(m_e q_e)^k K_{e,s}(\mathbf{x}) = c_e^k \frac{K_{e,h}(\mathbf{x})}{D_e(h)^k} + o_h(1), \quad (54)$$

uniformly on this range, in the ordered sense $\lim_{h \downarrow 0} \limsup_{e \downarrow 0} |o_h(1)| = 0$. Here and below a kernel includes its specified signed weights. To check the factors in this iteration, divide the single replacement estimate by $D_e(h) \geq c_0 a_{e,h}$. Each already replaced location contributes at most $S_e/D_e(h) \leq C a_e/a_{e,h}$, while its square kernel costs $C a_{e,h}$. Thus every accumulated error is $o_h(a_e^k)$ before normalization. The bound $m_e q_e a_e \leq C$ proves (54), including when c_e is small.

For fixed sufficiently small h , Proposition 4.5 gives common critical limits for $K_{e,h}$ and $D_e(h)$. The limiting denominator is positive: the square lower bound and the four-arm lower power bound give $D_e(h) \geq c_h > 0$ at this fixed h . These limits hold also for convergent insertion locations and are continuous on the separated range. Consequently the convergence is uniform there. Indeed, a failure of uniform convergence would give a sequence of locations with a convergent subsequence, contradicting the assertion for convergent locations. Since σ_e converges to Lebesgue measure, integrals against a fixed continuous cutoff therefore converge too. This applies to the triangular counting measure as well as to the Poisson location integral.

Apply this observation first with $k = 1$, $F = \mathbf{1}_{C_{Q_0}}$, and weight $\delta_1 - \delta_0$. Its full normalized first coefficient is

$$m_e q_e \int K_{e,s}(x) d\sigma_e(x) = q_e \mathbb{E} N_e(Q_0) = 1. \quad (55)$$

One may first use the localized box from Section 5; its omission contributes $o(1)$ also to this coefficient. By Lemma 5.2, for every $z > 0$ there is a continuous cutoff $0 \leq \chi \leq 1$, supported on a compact range away from the polygon boundary and the box boundary, such that the integral in (55) changes by at most $z + o(1)$ when multiplied by χ . Nonnegativity is used here.

Let $J_\chi(h)$ be the common limit of

$$\int \chi(x) \frac{K_{e,h}(x)}{D_e(h)} d\sigma_e(x).$$

The separate insertion upper bound and the square lower bound imply $0 \leq J_\chi(h) \leq C_\chi$, uniformly for sufficiently small h . For any subsequential limit c_* of c_e ,

$$|1 - c_* J_\chi(h)| \leq z + o_h(1). \quad (56)$$

Choose $z < 1/4$ and then h small. This proves that every such c_* is bounded below by a positive constant. If c_+ and c_- are two subsequential limits, possibly from different models, the same cutoff and the same $J_\chi(h)$ work for both. Since all c_e have a common upper bound C , (56) also gives $J_\chi(h) \geq (1 - z - o_h(1))/C$. Subtracting the two instances yields

$$|c_+ - c_-| \leq \frac{2C(z + o_h(1))}{1 - z - o_h(1)}.$$

First let $h \downarrow 0$ and then $z \downarrow 0$. All subsequential limits agree, proving $c_e \rightarrow c > 0$.

For a general fixed coefficient, restrict its location integral to a compact separated range, losing at most an arbitrarily small amount by Lemma 5.2. Equation (54) and the common critical square limits compare any two subsequential limits of the restricted coefficient. Their difference tends to zero as $h \downarrow 0$. Removing the cutoff proves uniqueness of the full coefficient limit across both models. Boundedness follows from Proposition 5.1, so these limits exist. The factor $1/k!$ in the expansion and the finite sum over replica assignments do not affect this argument. $\square$

This calibration uses a common macroscopic test, rather than a comparison of the microscopic scales at equal numerical mesh. We next convert it to the unit-square normalization of the theorem.

Lemma 7.2 (The prescribed normalization). *There is a common constant $B \in (0, \infty)$ such that, in each model,*

$$q_e \mathbb{E} N_e(Q_*) \longrightarrow B, \quad r_e = \frac{q_e}{B + o(1)}. \quad (57)$$

In particular, both prescribed scales tend to zero.

Proof. The first expression is the normalized first coefficient of $\mathbf{1}_{C_{Q_*}}$ with weight $\delta_1 - \delta_0$. Lemma 7.1 gives a common finite limit B . For positivity restrict insertion locations to a small open patch near the center of Q_* . The square lower bound of Lemma 6.1 gives $K_{e,h}(x) \geq c_1 a_{e,h}$ uniformly on a smaller patch, whereas $D_e(h) \leq C_1 a_{e,h}$. Choose a nonzero nonnegative cutoff in that patch. Equation (54) with $k = 1$ and $c_e \rightarrow c > 0$ shows that this part of the limiting first coefficient is strictly positive. All omitted parts are nonnegative. Hence $B > 0$. Taking reciprocals gives (57); $q_e \rightarrow 0$ by Lemma 2.12. $\square$

7.2 Finite tests and threshold laws

Proposition 7.3 (Common finite-time crossing limits). *Choose finitely many quads in $\mathcal{Q}_{\text{rat}}$ and finitely many real parameters. Their jointly coupled crossing indicators, with the prescribed scales r_e , have a limiting law common to the two models. The assertion holds for one color replica and for two independent color replicas conditional on the same geometry. Every such limiting probability is continuous in the real parameter vector, and the convergence remains valid for convergent parameter vectors.*

Proof. First use the auxiliary scale q_e . Expand the expectation of any bounded function of the specified indicators as in Section 5. The degree-zero term has its common critical limit by Proposition 4.5, and every positive-degree coefficient has a common limit by Lemma 7.1. The terms with repeated replica labels at one spatial site tend to zero. The majorant in Proposition 5.1 is summable uniformly on every bounded parameter range, so the whole expectation has a common limit.

At fixed degree the history weights are continuous on their finite labeled history space, including when times coincide or change order. The kernel integrals are multilinear in these weights. They therefore have continuous limiting coefficients, and local uniform summability makes their sum continuous. The same argument applies to convergent parameter vectors.

At the prescribed scale, the parameter λ corresponds to $t = \lambda/(B + o(1))$ at the auxiliary scale, by Lemma 7.2. The conclusion for convergent vectors proves the assertion in the required normalization. $\square$

Proof of Theorem 1.2. Finiteness and positivity of the expectations follow from Lemma 2.12; their inverse scales vanish by Lemma 7.2.

At fixed mesh, only finitely many cells meet any particular quad almost surely. Its crossing indicator is monotone in the parameter, and the closed mark rule gives, at every finite deterministic t ,

$$\{\tau_e(Q) \leq t\} = \{C_Q \text{ occurs at } p_e(t)\} \quad \text{almost surely.} \quad (58)$$

Thus Proposition 7.3 supplies common limits of the joint distribution functions of every finite set of thresholds. These limits are continuous at finite parameter vectors.

For completeness, compactification introduces no ambiguity. Take any weakly convergent subsequence of threshold laws on $\mathcal{K}$, which exists by compactness. At finite continuity points of a limiting coordinate marginal, (58) identifies its distribution function with the continuous limiting crossing function. Squeezing between such continuity points excludes a finite atom and gives the same identity at every finite point. Apply this to each coordinate of a finite projection. Its joint distribution function then agrees with the one supplied by Proposition 7.3. Finite-endpoint rectangles determine measures on $\overline{\mathbb{R}}^k$ as well: their increasing or decreasing limits recover events involving either endpoint at infinity. All finite projections are therefore uniquely determined. Since the index set is countable, they determine the law on $\mathcal{K}$.

In particular, the triangular laws have a full weak limit, which we denote by ν_Δ . The annealed one-replica Voronoi laws have the same limit. Apply the preceding argument also to two replicas. For the triangular model their joint law is a product at every mesh, so its limit is $\nu_\Delta \otimes \nu_\Delta$. Proposition 7.3 shows that two color replicas on a common Poisson geometry have this same limit.

Let f be a real continuous function on $\mathcal{K}$, and put

$$Z_{e,f} = \mathbb{E}[f(\tau_e^V) \mid \eta_e], \quad m_f = \int f \, d\nu_\Delta.$$

Conditional independence of the two color replicas gives

$$\mathbb{E}Z_{e,f} \longrightarrow m_f, \quad \mathbb{E}Z_{e,f}^2 = \mathbb{E}[f(\tau_e^{V,1})f(\tau_e^{V,2})] \longrightarrow m_f^2.$$

Hence $Z_{e,f} \rightarrow m_f$ in L^2 over geometry.

The family $\mathcal{F} = \{f : \|f\|_\infty \leq 1, \text{Lip}_{d_K}(f) \leq 1\}$ is uniformly bounded and equicontinuous on compact $\mathcal{K}$. For any $\zeta > 0$ it has a finite uniform ζ -net $f_1, \dots, f_m$. For every sampled geometry,

$$d_{\text{BL}}(\text{Law}(\tau_e^V \mid \eta_e), \nu_\Delta) \leq 2\zeta + \max_{1 \leq i \leq m} |Z_{e,f_i} - m_{f_i}|.$$

The maximum tends to zero in probability. Letting $\zeta \downarrow 0$ proves (6). All arguments apply along arbitrary vanishing-mesh sequences, completing the proof. $\square$

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